

these risks, and develop coping strategies that increase the chances of a return to abstinence if a slip occurs.

While many patients can be treated as outpatients, more intense interventions work better, and some alcoholics do not respond to AA or outpatient groups. Whatever the setting, subsequent contact with outpatient treatment staff should be maintained for a minimum of 6 months and preferably a full year after abstinence. Counseling focuses on areas of improved functioning in the absence of alcohol (i.e., why it is a good idea to continue to abstain) and helping the patient to manage free time without alcohol, develop a nondrinking peer group, and handle stresses on the job.

The physician serves an important role in identifying the alcoholic, diagnosing and treating associated medical or psychiatric syndromes, overseeing detoxification, referring the patient to rehabilitation programs, providing counseling, and, if appropriate, selecting which (if any) medication might be needed. For insomnia, patients should be reassured that troubled sleep is normal after alcohol withdrawal and will improve over subsequent weeks. They should be taught the basic elements of “sleep hygiene” including maintaining consistent schedules for bedtime and awakening. Sleep medications have the danger of being misused and of rebound insomnia when stopped. Sedating antidepressants (e.g., trazodone) should not be used as they interfere with cognitive functioning the next morning and disturb the normal sleep architecture, but occasional use of over-the-counter sleeping medications (sedating antihistamines) can be considered. Anxiety can be addressed by helping the person to gain insight into the temporary nature of the symptoms and to develop strategies to achieve relaxation as well as by using forms of cognitive therapy.

Medications for Rehabilitation Several medications have modest benefits when used for the first 6 months of recovery. The opioid-antagonist, naltrexone,

50–150 mg/d orally, appears to shorten subsequent relapses, whether used in the oral form or as a once-per-month 380-mg injection, especially in individuals with the G allele of the A118G polymorphism of the μ opioid receptor. By blocking opioid receptors, naltrexone may decrease activity in the dopamine-rich ventral tegmental reward system, and decrease the feeling of pleasure or reward if alcohol is imbibed. A second medication, acamprosate (Campral) at ~2 g/d divided into three oral doses, has similar modest effects; acamprosate inhibits NMDA receptors, decreasing mild symptoms of protracted withdrawal. Several trials of combined naltrexone and acamprosate using doses similar to those noted earlier have reported that the combination may be superior to either drug alone, although not all studies agree.

It is more difficult to establish the asset-to-liability ratio of a third drug, disulfiram, an ALDH inhibitor, used at doses of 250 mg/d. This drug produces vomiting and autonomic nervous system instability in the presence of alcohol as a result of rapidly rising blood levels of the first metabolite of alcohol, acetaldehyde. This reaction can be dangerous, especially for patients with heart disease, stroke, diabetes mellitus, or hypertension. The drug itself carries potential risks of depression, psychotic symptoms, peripheral neuropathy, and liver damage. Disulfiram is best given under supervision of another individual (such as a spouse), especially during discrete periods identified as representing high-risk drinking situations (such as the Christmas holiday). Other relevant drugs under investigation include the nicotinic receptor agonist varenicline, the serotonin antagonist ondansetron, the α -adrenergic agonist prazosin, the GABA-B receptor agonist baclofen, the anticonvulsant topiramate, and cannabinal receptor antagonists. At present, there are insufficient data to determine the asset-to-liability ratio for these medications in treating alcoholism and, therefore, no data to offer solid support for their use in clinical settings.

CHAPTER 57

OPIOID DRUG ABUSE AND DEPENDENCE

Thomas R. Kosten

INTRODUCTION

Opiate analgesics are some of the oldest and most common medications in clinical practice, but have also been abused since at least 300 B.C. Nepenthe (Greek “free from sorrow”) helped the hero of the *Odyssey*, but widespread opium smoking in China and the Near East has caused harm for centuries. Since the first chemical isolation of opium and codeine 200 years ago, a wide range of synthetic opioids have been developed, and endogenous opioid peptides were discovered in 1995. Two of the most important adverse effects of all these agents are overdose and dependence. The 0.14% annual prevalence of heroin dependence in the United States is only about one-third the rate of prescription opiate abuse and is substantially lower than the 2% rate of morphine dependence in Southeast and Southwest Asia. While these rates are low relative to other abused substances, their disease burden is substantial, with high rates of morbidity and mortality; disease transmission; increased health care, crime, and law enforcement costs; and less tangible costs of family distress and lost productivity.

The diagnosis of opiate dependence in the *Diagnostic and Statistical Manual of Mental Disorders*, Fourth Edition (DSM-IV) requires the repeated use of the drug while producing problems in three or more areas in a 12-month period. The areas include tolerance, withdrawal, use of greater amounts of opiates than intended, and use despite adverse consequences. The abuse diagnosis is related to legal problems, inability to meet obligations, use in hazardous situations, and continued use despite problems. The most striking aspect of opiate abuse has been its marked increase as the gateway to illicit drugs in the United States. Since 2007, prescription opiates have surpassed marijuana as the most common illicit drug that adolescents initially abuse.

The most commonly abused opiates are diverted prescriptions for oxycodone, followed by heroin and morphine, and—among health professionals—meperidine and

fentanyl. Two opiate maintenance treatment agents—methadone and buprenorphine—are also abused, but at substantially lower rates, and the partial opiate agonists such as butorphanol, tramadol, and pentazocine are infrequently abused. The chemistry and general pharmacology of these agents are covered in major pharmacology texts, and this chapter focuses on the neurobiology and pharmacology relevant to abuse, dependence, and their treatments.

NEUROBIOLOGY

During the past 30 years, substantial progress has been made in elucidating the neurobiology of opiates and their effects not only on the three types of opiate receptors (μ , κ , and δ) but also on the cascade of second, third, and fourth intracellular messenger systems and on neuronal action potentials. The different functional activities of these three receptors are summarized in [Table 57-1](#), and abuse liability is

TABLE 57-1

ACTIONS OF OPIOID RECEPTORS

RECEPTOR TYPE	ACTIONS
Mu (μ) (e.g., morphine)	Analgesia, reinforcement euphoria, cough and appetite suppression, decreased respirations, decreased GI motility, sedation, hormone changes, dopamine and acetylcholine release
Kappa (κ) (e.g., butorphanol)	Dysphoria, decreased GI motility, decreased appetite, decreased respiration, psychotic symptoms, sedation, diuresis, analgesia
Delta (δ) (e.g., etorphine)	Hormone changes, appetite suppression, dopamine release

Abbreviation: GI, gastrointestinal.

primarily associated with the mu receptor. A fourth type of opiate receptor, the orphanin receptor, also modulates pain but is not affected by opiate drugs. These opiate receptors are all G protein-linked and coupled to the cyclic adenosine monophosphate (cyclic AMP) second messenger system and to potassium channels. Opiates are inhibitory and block the potassium channels from opening and depolarizing the neuron, which would produce an action potential. Thus, opiates acutely inhibit neuronal activity. Analgesia and sedation are induced through this inhibition of specific brain pathways, while the “high” from opiates involves an indirect activation of a different brain pathway—the mesolimbic dopamine pathway.

The various effects of opiates are related to the specific neuroanatomic locations of mu receptors. Reinforcing and euphoric effects of opiates occur in the dopaminergic pathway from the ventral tegmental area (VTA) to the nucleus accumbens, where opiates increase synaptic levels of dopamine. This increase is due to inhibition of GABAergic neurons that inhibit both the VTA and nucleus accumbens activity. However, the “high” only occurs when the *rate of change* in dopamine is fast. Large, rapidly administered doses of opiates block GABA inhibition and produce a burst of nucleus accumbens activity that is associated with “high” in all abused drugs. Therefore, routes of administration that slowly increase opiate blood and brain levels, such as oral and transdermal routes, are effective for analgesia and sedation but do not produce an opiate “high” that follows smoking and intravenous routes. Other acute effects such as analgesia and respiratory depression leading to overdose are due to stimulation of opiate receptors located in other areas such as the locus coeruleus.

Opiate dependence and withdrawal are chronic effects related to the cyclic AMP system. This second messenger phosphorylates various intracellular proteins and produces a cascade of changes reaching into the nucleus and DNA. Immediate early gene products such as *c-fos* and *c-jun* are activated followed by regulation of other genes with more sustained protein transcription such as *delta c-fos*. With these sustained gene activations, several receptor-level changes occur, including downregulation of receptor numbers, reduced neuronal cell-surface receptor trafficking, uncoupling of G proteins from the mu opiate receptors, and upregulation of cyclic AMP second messenger systems. These effects are also reflective of genetic risk factors for drug dependence, with estimates of up to 50% of the risk for dependence due to polygenic inheritance. Specific functional genetic polymorphisms in the mu opiate receptor gene appear associated with this risk for opiate abuse, including one producing a threefold increase in this receptor’s affinity for opiates and the

endogenous ligand beta endorphin. Epigenetic methylation changes also occur on the DNA of the mu receptor gene of opiate addicts. DNA methylation inhibits gene transcription.

This molecular cascade links acute intoxication and sedation to chronic opiate dependence and withdrawal within the specific neuroanatomic structure of the locus coeruleus. The locus coeruleus is the brain’s largest concentration of noradrenergic neurons and is responsible for a large proportion of brain cortical activation. When large opiate doses saturate and activate all of its mu receptors, its steady rate of action potentials can cease due to the inactivation of potassium channels. When this direct inhibitory effect is sustained over weeks and months of opiate use, a secondary set of regulatory effects take place in the cyclic AMP system that leads to tolerance, dependence, and withdrawal symptoms.

Opiate withdrawal symptoms reflect overactivity of adrenergic neurons that are located in the locus coeruleus. Opiates suppress the activity of these neurons, and when this suppression continues chronically from daily opiate use, a secondary upregulation occurs in adenylyl cyclase enzyme capacity and the production of cyclic AMP from ATP. This upregulation is a homeostatic response to the chronic opiate suppression, but when that suppression is terminated by discontinuing the opiate, this enhanced adenylyl cyclase activity leads to a marked increase in cyclic AMP. The now very high levels of cyclic AMP activate the sodium-potassium channels and produce a high level of action potentials in these adrenergic neurons. This adrenergic arousal is one basis for the symptoms of opiate withdrawal and takes about 7 days to readjust to normal levels of adenylyl cyclase activity and the associated resolution of opiate withdrawal symptoms. This molecular model of adrenergic neuronal activation during withdrawal has had important treatment implications, such as the use of clonidine for opioid withdrawal.

PHARMACOLOGY

Tolerance and withdrawal commonly occur with chronic daily use as quickly as 6–8 weeks depending on the dose and frequency of dosing. Tolerance appears to be primarily a pharmacodynamic rather than pharmacokinetic effect, with relatively limited induction of the cytochrome P450 or other liver enzymes. The metabolism of opiates occurs in the liver primarily through the cytochrome P450 systems of 2D6 and 3A4. They then are conjugated to glucuronic acid and excreted in small amounts in feces. The plasma half-lives generally range from 2.5 to 3 h for morphine and more than 22 h for methadone. The shortest half-lives of several minutes are for fentanyl-related opiates and the longest are for buprenorphine and its active

metabolites, which can block opiate withdrawal for up to 3 days after a single dose. Tolerance to the mental effects of opioids leads to the need for ever-increasing amounts of drugs to sustain the desired euphoriant effects—as well as to avoid the discomfort of withdrawal. This combination has the expected consequence of strongly reinforcing dependence once it has started. The role of endogenous opioid peptides in opioid dependence is uncertain.

The clinical aspects of abuse are tied to route of administration and the rapidity of an opiate bolus in reaching the brain. Intravenous and smoked administration is routine not only because it is the most efficient route but also because it rapidly produces a bolus of high drug concentration in the brain. This bolus produces a “rush,” followed by euphoria, a feeling of tranquility, and sleepiness (“the nod”). Heroin produces effects that last 3–5 h, and several doses a day are required to forestall manifestations of withdrawal in dependent persons. Symptoms of opioid withdrawal begin 8–10 h after the last dose. Many of these symptoms reflect increased activity of the autonomic nervous system. Lacrimation, rhinorrhea, yawning, and sweating appear first. Restless sleep followed by weakness, chills, gooseflesh (“cold turkey”), nausea and vomiting, muscle aches, and involuntary movements (“kicking the habit”), hyperpnea, hyperthermia, and hypertension occur in later stages of the withdrawal syndrome. The acute course of withdrawal may last 7–10 days. A secondary phase of protracted abstinence lasts for 26–30 weeks and is characterized by hypotension, bradycardia, hypothermia, mydriasis, and decreased responsiveness of the respiratory center to carbon dioxide.

Opioid effects on organs

Besides the brain effects of opioids on sedation and euphoria and the combined brain and peripheral nervous system effects on analgesia, a wide range of other organs can be affected. The cough reflex is inhibited through the brain, leading to the use of some opiates as an antitussive, and nausea and vomiting are due to brainstem effects on the medulla. The release of several hormones is inhibited including corticotropin-releasing factor (CRF) and luteinizing hormone, which reduce cortisol and sex hormone levels, respectively. The clinical manifestations of these reductions can involve poor responses to stress and reductions in sex drive. An increase in prolactin also contributes to the reduced sex drive in males. Two other hormones affected are decreased thyrotropin and increased growth hormone. Respiratory depression results from opiate-induced insensitivity of brainstem neurons to increases in carbon dioxide. This depression contributes to overdose, but in patients with pulmonary disease even opiate doses

well below those typical of overdose can produce clinically significant complications. In overdoses, aspiration pneumonia is a common complication due to loss of the choking reflex. Opiates reduce gut motility, which is helpful for diarrhea, but can lead to nausea, constipation, and anorexia with weight loss. Deaths have occurred in early methadone maintenance programs due to severe constipation and toxic megacolon. Opiates may prolong QT intervals and lead to sudden death in some patients. Two opiates particularly noted for this complication are methadone and a long-acting form of methadone called LAAM that was withdrawn from the market. Orthostatic hypotension may occur due to histamine release and peripheral blood vessel dilation, which is an opiate effect usefully applied to managing acute myocardial infarction.

Heroin users in particular tend to use opiates intravenously and be polydrug users, also using alcohol, sedatives, cannabinoids, and stimulants. None of these other drugs serve as substitutes for opioids, but they have desired additive effects. One needs to be sure that the person undergoing a withdrawal reaction is not also withdrawing from alcohol or sedatives, which might be more dangerous and more difficult to manage.

Besides the ever-present risk of fatal overdose, hepatitis B and AIDS are among the many potential complications of sharing contaminated hypodermic syringes. Bacterial infections lead to septic complications such as meningitis, osteomyelitis, and abscesses in various organs. Attempts to illicitly manufacture meperidine in the 1980s resulted in the production of a highly specific neurotoxin, MPTP, which produced parkinsonism in users.

Toxicity and overdose

Lethal overdose is a relatively common complication of opiate dependence and must be rapidly recognized and treated because naloxone provides a highly specific reversal agent that is relatively free of complications. The diagnosis generally does not rely on blood or urine toxicology results but on clinical signs and symptoms. The presentation involves shallow and slow respirations, pupillary miosis (mydriasis does not occur until significant brain anoxia supervenes), bradycardia, hypothermia, and stupor or coma. If naloxone is not administered, progression to respiratory and cardiovascular collapse leading to death occurs. At autopsy, cerebral edema and sometimes frothy pulmonary edema are generally found, but those pulmonary effects are most likely from allergic reactions to adulterants mixed with the heroin. Opiates generally do not produce seizures except for unusual cases of mixed drug abuse with the opiate meperidine or with high doses of tramadol.

TREATMENT **Opioid Overdose**

Beyond the acute treatment of opiate overdose with naloxone, clinicians have two general treatment paths: opioid maintenance treatment or detoxification. Most opioid-dependent individuals engage in multiple episodes of all three categories of treatment during the course of their drug-using careers. Agonist and partial agonist medications are commonly utilized for both maintenance and detoxification purposes. Alpha-2-adrenergic agonists are primarily used for detoxification. Antagonists are used to accelerate detoxification and then continued postdetoxification to prevent relapse. Only the residential medication-free programs have had success that comes close to matching that of the medication-based programs. Success of the various treatment approaches is assessed as retention in treatment, reduced opioid and other drug use, as well as secondary outcomes such as HIV risk behaviors, crime, psychiatric symptoms, and medical comorbidity.

Stopping opiates is like stopping most drugs of abuse—it is much easier to stop than to prevent relapses. Long-term relapse prevention for opioid-dependent persons requires combined pharmacologic and psychosocial approaches. Chronic users tend to prefer pharmacologic approaches; those with shorter histories of drug abuse are more amenable to detoxification and psychosocial interventions.

Opiate Overdose Treatment Managing overdose requires naloxone and support of vital functions, including intubation if needed. The opiate antagonist naloxone is given at 0.4–2 mg IV or IM, with an expected response within 1–2 min. If the overdose is due to buprenorphine, then naloxone might be required at total doses of 10 mg or greater, but primary buprenorphine overdose is nearly impossible because this agent is a partial opiate agonist. Partial agonism means that as the dose of buprenorphine is increased, it has greater opiate antagonist than agonist activity. Thus, a 0.2-mg buprenorphine dose leads to analgesia and sedation, while a hundred times greater 20-mg dose produces profound opiate antagonism, precipitating opiate withdrawal in a person who was opiate dependent on morphine or methadone. When 10 mg of naloxone fails to produce arousal in the patient, another cause of toxicity must be found. Before reaching such large naloxone doses, however, it is important to recognize that the goal is to reverse the respiratory depression and not to administer so much naloxone that it precipitates opiate withdrawal. Because naloxone only lasts a few hours and most opiates last considerably longer, close monitoring and an IV naloxone drip is frequently employed to provide a continuous level of antagonism for 24–72 h depending on the opiate used in the overdose (e.g.,

morphine vs. methadone). Other sedative drugs that produce significant overdoses must also be considered if naloxone has only a limited effect. The most common are benzodiazepines, which have produced overdoses and deaths in combination with buprenorphine. A specific antagonist for benzodiazepines—flumazenil at 0.2 mg/min—can be given to a maximum of 3 g/h, but it may precipitate seizures and increase intracranial pressure. Like naloxone, administration for a prolonged period is usually required since most benzodiazepines remain active for considerably longer than flumazenil.

Support of vital functions may include oxygen and positive-pressure breathing, IV fluids, pressor agents for hypotension, and cardiac monitoring to detect QT prolongation, which might require treatment. Activated charcoal and gastric lavage may be helpful for oral ingestions, but intubation will be needed if the patient is stuporous.

Opiate Withdrawal Treatment The principles of detoxification are the same for all drugs: to substitute a longer-acting, orally active, pharmacologically equivalent drug for the abused drug, stabilize the patient on that drug, and then gradually withdraw the substituted drug. Methadone is admirably suited for such use in opioid-dependent persons, and the partial mu agonist buprenorphine is another option. Clonidine, a centrally acting sympatholytic agent, has also been used for detoxification. By reducing central sympathetic outflow, clonidine mitigates many of the signs of sympathetic overactivity. Clonidine has no narcotic action and is not addictive. Lofexidine, a clonidine analogue with less hypotensive effect, is being developed for use.

Methadone for Detoxification Methadone dose tapering regimens for detoxification range from 2 to 3 weeks to as long as 180 days, but this approach is controversial given the relative effectiveness of methadone maintenance and the low success rates of detoxification. Unfortunately, the vast majority of patients tend to relapse to heroin or other opiates during or after the detoxification period, indicative of the chronic and relapsing nature of opioid dependence.

Buprenorphine for Detoxification Because it is a partial agonist, buprenorphine produces fewer withdrawal symptoms and may allow briefer detoxifications compared with full agonists like methadone, but it does not appear to have better outcomes than methadone tapering. Buprenorphine is superior to the alpha-2-adrenergic agonist clonidine in reducing symptoms of withdrawal, retaining patients in a withdrawal protocol, and in treatment completion.

Alpha-2-Adrenergic Agonists for Detoxification Several alpha-2-adrenergic agonists have relieved opioid withdrawal by suppressing central

noradrenergic hyperactivity. Alpha-2-adrenergic agonists moderate the symptoms of noradrenergic hyperactivity via actions in the central nervous system. Clonidine relieves some signs and symptoms of opiate withdrawal such as lacrimation, rhinorrhea, muscle pain, joint pain, restlessness, and gastrointestinal symptoms, but it is not a drug of abuse or dependence. Unfortunately, clonidine is associated with significant hypotension, which has stimulated investigation of lofexidine, guanfacine, and guanabenz acetate. Lofexidine can be dosed up to ~2 mg/d and appears to be associated with fewer adverse effects, and it is therefore likely to replace clonidine as the leading opioid withdrawal treatment in this drug class. Clonidine or lofexidine are typically administered orally, in three or four doses per day, with dizziness, sedation, lethargy, and dry mouth as the primary adverse side effects. Completion rates of managed withdrawal assisted with clonidine and other alpha-2-adrenergic agents vs. methadone have been comparable.

Rapid and Ultrarapid Opiate Detoxification

The opioid antagonist naltrexone typically combined with an alpha-2-adrenergic agonist has been purported to shorten the duration of withdrawal without significantly increasing patient discomfort. Another benefit to rapid opiate detoxification (ROD) is the reduced time between opioid use and the commencement of sustained naltrexone treatment for prevention of relapse (discussed later). ROD completion rates using naltrexone and clonidine range from 75 to 81% compared to 40 to 65% for methadone or clonidine alone. Buprenorphine in combination with naltrexone and clonidine reduced ROD from 3 days to 1 day of detoxification. Ultrarapid opiate detoxification is an extension of ROD using anesthetics, but is highly controversial due to the medical risks and mortality associated with it.

Agonist medications for opioid dependence

Methadone maintenance substitutes a once-daily oral opioid dose for three-to-four times daily heroin. Methadone saturates the opioid receptors, and by inducing a high level of opiate tolerance, blocks the desired euphoria from additional opiates. Buprenorphine, a partial opioid agonist, also can be given once daily at sublingual doses of 4–32 mg daily, and in contrast to methadone it can be given in an office-based primary care setting.

Methadone maintenance

Methadone's slow onset of action when taken orally, long elimination half-life (24–36 h), and production of cross-tolerance at doses from 80 to 150 mg are the basis for its efficacy in treatment retention and reductions in

IV drug use, criminal activity, and HIV risk behaviors and mortality. Methadone can prolong the QT interval at rates as high as 16% above the rates in non-methadone-maintained, drug-injecting patients, but it has been used safely in the treatment of opioid dependence for 40 years.

Buprenorphine maintenance

While France and Australia have had sublingual buprenorphine maintenance since 1996, the USFDA approved it as a Schedule III drug in 2002 for managing opiate dependence. Unlike the full agonist methadone, buprenorphine is a partial agonist of mu-opioid receptors with a slow onset and long duration of action, allowing for alternate-day dosing. Its partial agonism reduces the risk of unintentional overdose but limits its efficacy to patients who need the equivalent of only 60–70 mg of methadone, and many patients in methadone maintenance require higher doses up to 150 mg daily. Buprenorphine is combined with naloxone at a 4:1 ratio in order to reduce its abuse liability. A subcutaneous buprenorphine implant has also been tested, but results are not yet available.

In the United States, the ability of primary care physicians to prescribe buprenorphine for opioid dependence presents an important and far-reaching opportunity to improve access and quality of treatment as well as reduce social harm. Europe, Asia, and Australia have found reduced opioid-related deaths and drug-injection-related medical morbidity with buprenorphine available in primary care. Retention in office-based buprenorphine treatment has been greater than 70% at 6-month follow-ups.

Antagonist medications for opioid dependence

The rationale for using narcotic antagonist therapy is that blocking the action of self-administered opioids should eventually extinguish the habit, but this therapy is poorly accepted by patients. Naltrexone, a long-acting orally active pure opioid antagonist, can be given three times a week at doses of 100–150 mg and a depot form for monthly administration is available. Because it is an antagonist, the patient must first be detoxified from opioid dependence before starting naltrexone. When taken chronically for even years, it is safe, associated with few side effects (headache, nausea, abdominal pain), and can be given to patients infected with hepatitis B or C without producing hepatotoxicity. However, most providers refrain from prescribing it if liver function tests are 3–5 times above normal levels. Naltrexone maintenance combined with psychosocial therapy is effective in reducing heroin use, but medication adherence is low. Depot injection formulations lasting up to 4 weeks markedly improve adherence, retention, and drug use. Subcutaneous naltrexone implants in Russia,

Medication-free treatment

Most opiate addicts enter medication-free treatments in inpatient, residential, or outpatient settings, but 1- to 5-year outcomes are very poor compared to pharmacotherapy except for residential settings lasting 6 to 18 months. The residential programs require full immersion in a regimented system that has progressively increasing levels of independence and responsibility within a controlled community of fellow drug abusers. These medication-free programs, as well as the pharmacotherapy programs, also include counseling and behavioral treatments designed to teach interpersonal and cognitive skills for coping with stress and for avoiding situations leading to easy access to drugs or to craving. Relapse is prevented by having the individual very gradually reintroduced to greater responsibilities and to

the working environment outside of the protected therapeutic community.

PREVENTION

Preventing opiate abuse represents a critically important challenge for physicians. Opiate prescriptions are the most common source of drugs accessed by adolescents who begin a pattern of illicit drug abuse; in the United States, 9000 adolescents become opiate abusers every day. The major sources of these drugs are family members, not drug dealers or the Internet. Pain management involves giving sufficient opiates to relieve the pain over as short a period of time as the pain warrants. The patient then needs to dispose of any remaining opiates, not save them in the medicine cabinet, because this behavior leads to diversion to adolescents. Finally, physicians should never prescribe opiates for themselves.

CHAPTER 58

COCAINE AND OTHER COMMONLY ABUSED DRUGS



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The abuse of cocaine and other psychostimulant drugs reflects a complex interaction between the pharmacologic properties of each drug, the personality and expectations of the user, and the environmental context in which the drug is used. Polydrug abuse involving the concurrent use of several drugs with different pharmacologic effects is increasingly common. Some forms of polydrug abuse, such as the combined use of heroin and cocaine intravenously, are especially dangerous and are a major problem in hospital emergency rooms. Sometimes one drug is used to enhance the effects of another, as with the combined use of benzodiazepines and methadone, or cocaine and heroin in methadone-maintained patients.

Chronic cocaine and psychostimulant abuse may cause a number of adverse health consequences, and preexisting disorders such as hypertension and cardiac disease may be exacerbated by drug abuse. The combined use of two or more drugs may accentuate medical complications associated with abuse of one of them. Chronic drug abuse is often associated with immune system dysfunction and increased vulnerability to infections, which in turn contributes to the risk for HIV infection. In addition, concurrent use of cocaine and opiates (the “speedball”) is frequently associated with needle sharing by IV drug users. Intravenous drug abusers continue to represent the largest single group of persons with HIV infection in several major metropolitan areas in the United States as well as in many parts of Europe and Asia.

COCAINE

Cocaine is a stimulant and a local anesthetic with potent vasoconstrictor properties. The leaves of the *coca* plant (*Erythroxylon coca*) contain ~0.5–1% cocaine. The drug

produces physiologic and behavioral effects when administered orally, intranasally, intravenously, or via inhalation following pyrolysis (smoking). The reinforcing effects of cocaine appear to be related to activation of dopaminergic neurons in the mesolimbic system. Cocaine increases synaptic concentrations of the monoamine neurotransmitters dopamine, norepinephrine, and serotonin by binding to transporter proteins in presynaptic neurons and blocking reuptake.

Prevalence of cocaine use

Cocaine is widely available throughout the United States, and cocaine abuse occurs in virtually all social and economic strata of society. The prevalence of cocaine abuse in the general population has been accompanied by an increase in cocaine abuse by heroin-dependent persons, including those in methadone maintenance programs. Intravenous cocaine is often used concurrently with IV heroin. This combination purportedly attenuates the postcocaine “crash” and substitutes a cocaine “high” for the heroin “high” blocked by methadone.

Acute and chronic intoxication

There has been an increase in both IV administration and inhalation of pyrolyzed cocaine via smoking. Following intranasal administration, changes in mood and sensation are perceived within 3–5 min, and peak effects occur at 10–20 min. The effects rarely last more than 1 h. Inhalation of pyrolyzed materials includes inhaling crack/cocaine or smoking coca paste, a product made by extracting cocaine preparations with flammable solvents, and cocaine free-base smoking. Free-base cocaine, including the freebase prepared with sodium bicarbonate (crack), has become increasingly

^aDeceased.

popular because of the relative high potency of the compound and its rapid onset of action (8–10 s following smoking).

Cocaine produces a brief, dose-related stimulation and enhancement of mood and an increase in cardiac rate and blood pressure. Body temperature usually increases following cocaine administration, and high doses of cocaine may induce lethal pyrexia or hypertension. Because cocaine inhibits reuptake of catecholamines at adrenergic nerve endings, the drug potentiates sympathetic nervous system activity. Cocaine has a short plasma half-life of approximately 45–60 min. It is metabolized by plasma esterases, and cocaine metabolites are excreted in urine. The very short duration of the euphorigenic effects of cocaine observed in chronic abusers is probably due to both acute and chronic tolerance. Frequent self-administration of the drug (two to three times per hour) is often reported by chronic cocaine abusers. Alcohol is often used to modulate both the cocaine high and the dysphoria associated with the abrupt disappearance of cocaine's effects. A metabolite of cocaine, cocaethylene, has been detected in the blood and urine of persons who concurrently abuse alcohol and cocaine. Cocaethylene induces changes in cardiovascular function similar to those of cocaine alone, and the pathophysiologic consequences of alcohol abuse plus cocaine abuse may be additive when both are used together.

The prevalent assumption that cocaine inhalation or IV administration is relatively safe is contradicted by reports of death from respiratory depression, cardiac arrhythmias, and convulsions associated with cocaine use. In addition to generalized seizures, neurologic complications may include headache, ischemic or hemorrhagic stroke, or subarachnoid hemorrhage. Disorders of cerebral blood flow and perfusion in cocaine-dependent persons have been detected with magnetic resonance spectroscopy (MRS) studies. Severe pulmonary disease may develop in individuals who inhale crack cocaine; this effect is attributed both to the direct effects of cocaine and to residual contaminants in the smoked material. Hepatic necrosis may occur following chronic crack/cocaine use. Protracted cocaine abuse may also cause paranoid ideation and visual and auditory hallucinations, a state that resembles alcoholic hallucinosis.

Although men and women who abuse cocaine may report that the drug enhances libidinal drive, chronic cocaine use causes significant loss of libido and adversely affects sexual function. Impotence and gynecomastia have been observed in male cocaine abusers, and these abnormalities often persist for long periods following cessation of drug use. Women who abuse cocaine may have major derangements in menstrual cycle function, including galactorrhea, amenorrhea, and infertility. Chronic cocaine abuse may cause persistent hyperprolactinemia as a consequence of disordered dopaminergic

inhibition of prolactin secretion by the anterior pituitary. Cocaine abuse by pregnant women, particularly crack smoking, has been associated with both an increased risk of congenital malformations in the fetus and perinatal cardiovascular and cerebrovascular disease in the mother. However, cocaine abuse per se is probably not the sole cause of these perinatal disorders, because maternal cocaine abuse is often associated with poor nutrition and prenatal health care as well as polydrug abuse that may contribute to the risk for perinatal disease.

Psychological dependence on cocaine, indicated by inability to abstain from frequent compulsive use, has also been reported. Although the occurrence of withdrawal syndromes involving psychomotor agitation and autonomic hyperactivity remains controversial, severe depression ("crashing") following cocaine intoxication may accompany drug withdrawal.

TREATMENT

Cocaine Overdose and Chronic Abuse

Treatment of cocaine overdose is a medical emergency that is best managed in an intensive care unit. Cocaine toxicity produces a hyperadrenergic state characterized by hypertension, tachycardia, tonic-clonic seizures, dyspnea, and ventricular arrhythmias. Intravenous diazepam in doses up to 0.5 mg/kg administered over an 8-h period has been shown to be effective for control of seizures. Ventricular arrhythmias have been managed successfully by administration of 0.5–1 mg of propranolol IV. Since many instances of cocaine-related mortality have been associated with concurrent use of other illicit drugs (particularly heroin), the physician must be prepared to institute effective emergency treatment for multiple drug toxicities.

Treatment of chronic cocaine abuse requires the combined efforts of primary care physicians, psychiatrists, and psychosocial care providers. Early abstinence from cocaine use is often complicated by symptoms of depression and guilt, insomnia, and anorexia, which may be as severe as those observed in major affective disorders. Individual and group psychotherapy, family therapy, and peer group assistance programs are often useful for inducing prolonged remission from drug use. A number of medications used for the treatment of various medical and psychiatric disorders have been administered to reduce the duration and severity of cocaine abuse and dependence. The search for a medication that is both safe and highly effective for cocaine detoxification or maintenance of abstinence is continuing. Although psychotherapy may be effective, no specific form of psychotherapy or behavioral modification is uniquely beneficial.

MARIJUANA AND CANNABIS COMPOUNDS

Cannabis sativa contains >400 compounds in addition to the psychoactive substance, delta-9-tetrahydrocannabinol (THC). Marijuana cigarettes are prepared from the leaves and flowering tops of the plant, and a typical marijuana cigarette contains 0.5–1 g of plant material. The usual THC concentration varies between 10 and 40 mg, but concentrations >100 mg per cigarette have been detected. Hashish is prepared from concentrated resin of *C. sativa* and contains a THC concentration of between 8 and 12% by weight. “Hash oil,” a lipid-soluble plant extract, may contain THC between 25 and 60% and may be added to marijuana or hashish to enhance its THC concentration. Smoking is the most common mode of marijuana or hashish use. During pyrolysis, >150 compounds in addition to THC are released in the smoke. Although most of these compounds do not have psychoactive properties, they may have physiologic effects.

THC is quickly absorbed from the lungs into blood, and then rapidly sequestered in tissues. THC is metabolized primarily in the liver, where it is converted to 11-hydroxy-THC, a psychoactive compound, and >20 other metabolites. Many THC metabolites are excreted through the feces at a relatively slow rate of clearance in comparison to most other psychoactive drugs.

Specific cannabinoid receptors (CB₁ and CB₂) have been identified in the central and peripheral nervous system. High densities of cannabinoid receptors have been found in the cerebral cortex, basal ganglia, and hippocampus. T and B lymphocytes also contain cannabinoid receptors, and these appear to mediate the anti-inflammatory and immunoregulatory properties of cannabinoids. A naturally occurring THC-like ligand has been identified and is widely distributed in the nervous system.

Prevalence of use

Marijuana is the most commonly used illegal drug in the United States, and its use is particularly prevalent among adolescents. Marijuana is relatively inexpensive and is often considered to be less hazardous than other controlled drugs and substances. Very potent forms of marijuana (sinsemilla) are now available in many locations, and concurrent use of marijuana with crack/cocaine and phencyclidine is not uncommon.

Acute and chronic intoxication

Acute intoxication from marijuana and cannabis compounds is related to both the dose of THC and the route of administration. THC is absorbed more rapidly

from marijuana smoking than from orally ingested cannabis compounds. Acute marijuana intoxication may produce a perception of relaxation and mild euphoria resembling mild to moderate alcohol intoxication. This condition is usually accompanied by some impairment in thinking, concentration, and perceptual and psychomotor function. Higher doses of cannabis may produce more pronounced impairment in concentration and perception, as well as greater sedation. Although the acute effects of marijuana intoxication are relatively benign in normal users, the drug can precipitate severe emotional disorders in individuals who have antecedent psychotic or neurotic problems. Like other psychoactive compounds, both the user's expectations and the environmental context are important determinants of the type and severity of the effects of marijuana intoxication.

As with abuse of cocaine, opioids, and alcohol, chronic marijuana abusers may lose interest in common socially desirable goals and steadily devote more time to drug acquisition and use. However, THC does not cause a specific and unique “amotivational syndrome.” The range of symptoms sometimes attributed to marijuana use is difficult to distinguish from mild to moderate depression and the maturational dysfunctions often associated with protracted adolescence. Chronic marijuana use has also been reported to increase the risk of psychotic symptoms in individuals with a past history of schizophrenia. Persons who initiate marijuana smoking before the age of 17 may exhibit more pronounced cognitive deficits and they may also be at higher risk for polydrug and alcohol abuse problems in later life, but the role of marijuana in this causal sequence is uncertain.

Physical effects

Conjunctival injection and tachycardia are the most frequent immediate physical concomitants of smoking marijuana. Tolerance for marijuana-induced tachycardia develops rapidly among regular users. However, marijuana smoking may precipitate angina in persons with a history of coronary insufficiency. Exercise-induced angina may be increased after marijuana use to a greater extent than after tobacco cigarette smoking. Patients with cardiac disease should be strongly advised not to smoke marijuana or use cannabis compounds.

Significant decrements in pulmonary vital capacity have been found in regular daily marijuana smokers. Because marijuana smoking typically involves deep inhalation and prolonged retention of marijuana smoke, marijuana smokers may develop chronic bronchial irritation. Impairment of single-breath carbon monoxide diffusion capacity (DL_{CO}) is greater in persons who smoke both marijuana and tobacco than in tobacco smokers.

Although marijuana has also been associated with a number of other adverse effects, many of these studies await replication and confirmation. A reported correlation between chronic marijuana use and decreased testosterone levels in males has not been confirmed. Decreased sperm count and sperm motility and morphologic abnormalities of spermatozoa following marijuana use have been reported. Prospective studies demonstrated a correlation between impaired fetal growth and development and heavy marijuana use during pregnancy. Marijuana has also been implicated in derangements of the immune system; in chromosomal abnormalities; and in inhibition of DNA, RNA, and protein synthesis; however, these findings have not been confirmed or related to any specific physiologic effect in humans.

Tolerance and physical dependence

Habitual marijuana users rapidly develop tolerance to the psychoactive effects of marijuana, then smoke more frequently and try to acquire more potent cannabis compounds. Tolerance for the physiologic effects of marijuana develops at different rates; e.g., tolerance develops rapidly for marijuana-induced tachycardia but more slowly for marijuana-induced conjunctival injection. Tolerance for both behavioral and physiologic effects of marijuana decreases rapidly upon cessation of marijuana use.

Withdrawal signs and symptoms have been reported in chronic cannabis users, and the severity of symptoms is related to dosage and duration of use. These symptoms typically reach their peak several days after cessation of chronic use and include irritability, anorexia, and sleep disturbances. Withdrawal signs and symptoms observed in chronic marijuana users are usually relatively mild in comparison to those observed in heavy opiate or alcohol users and rarely require medical or pharmacologic intervention. However, more severe and protracted abstinence syndromes may occur after sustained use of high-potency cannabis compounds.

Therapeutic use of marijuana

Marijuana, administered as cigarettes or as a synthetic oral cannabinoid (dronabinol), has been proposed to have a number of medicinal properties that may be clinically useful in some situations. These include antiemetic effects in chemotherapy recipients, appetite-promoting effects in AIDS patients, reduction of intraocular pressure in glaucoma, and reduction of spasticity in multiple sclerosis and other neurologic disorders. With the possible exception of AIDS-related cachexia, none of these attributes of marijuana compounds is clearly superior to other readily available therapies.

METHAMPHETAMINE

Methamphetamine is also referred to as “meth,” “speed,” “crank,” “chalk,” “ice,” “glass,” or “crystal.” Methamphetamine is a mixed-action monoamine releaser with activity at dopamine, serotonin, and norepinephrine systems. Despite drug seizures, closures of clandestine laboratories that produce methamphetamine illegally, and an increase in methamphetamine abuse prevention programs, methamphetamine was considered second only to cocaine as a drug threat to society by the U.S. Department of Justice in 2009. Hospital admissions for methamphetamine treatment more than doubled between 1998 and 2007, and young adults (aged 18–25 years) have the highest use rates.

Methamphetamine can be used by smoking, snorting, IV injection, or oral administration. Methamphetamine abusers report that drug use induces feelings of euphoria and decreased fatigue. Adverse consequences of methamphetamine abuse include headache, difficulty concentrating, diminished appetite, abdominal pain, vomiting or diarrhea, disordered sleep, paranoid or aggressive behavior, and psychosis. Chronic methamphetamine abuse can result in severe dental caries, described as blackened, rotting, crumbling teeth. Severe, life-threatening methamphetamine toxicity may include hypertension, cardiac arrhythmia or cardiac failure, subarachnoid hemorrhage, ischemic stroke, intracerebral hemorrhage, convulsions, or coma. Methamphetamines increase the release of monoamine neurotransmitters (dopamine, norepinephrine, and serotonin) from presynaptic neurons. It is thought that the euphoric and reinforcing effects of this class of drugs are mediated through dopamine and the mesolimbic system, whereas the cardiovascular effects are related to norepinephrine. MRS studies of the brain suggest that chronic abusers have neuronal damage in the frontal areas and basal ganglia.

Therapy of acute methamphetamine overdose is largely symptomatic. Ammonium chloride may be useful to acidify the urine and enhance clearance of the drug. Hypertension may respond to sodium nitropruside or α -adrenergic antagonists. Sedatives may reduce agitation and other signs of central nervous system hyperactivity. Treatment of chronic methamphetamine dependence may be accomplished in either an inpatient or outpatient setting using strategies similar to those described earlier for cocaine abuse.

MDMA (3,4-methylenedioxymethamphetamine), or *ecstasy*, is a derivative of methamphetamine. Ecstasy is usually taken orally but may be injected or inhaled; its effects last for 3–6 h. In addition to amphetamine-like effects, MDMA can induce hyperthermia and vivid hallucinations and other perceptual distortions. Recent studies have revealed that MDMA use is associated with cognitive and

memory impairment and a mild withdrawal syndrome after cessation of use. The long-term consequences of recreational use of MDMA by young persons are poorly understood.

LYSERGIC ACID DIETHYLAMIDE (LSD)

The discovery of the psychedelic effects of LSD led to an epidemic of LSD abuse during the 1960s. Imposition of stringent constraints on the manufacture and distribution of LSD (classified as a Schedule I substance by the U.S. Food and Drug Administration), as well as public recognition that psychedelic experiences induced by LSD were a health hazard, have resulted in a reduction in LSD abuse. LSD still remains popular among adolescents and young adults, and there are indications that LSD use among young persons has been increasing in some areas in the United States.

LSD is a very potent hallucinogen; oral doses as low as 20 µg may induce profound psychological and physiologic effects. Tachycardia, hypertension, pupillary dilation, tremor, and hyperpyrexia occur within minutes following oral administration of 0.5–2 µg/kg. A variety of bizarre and often conflicting perceptual and mood changes, including visual illusions, synesthesias, and extreme lability of mood, usually occur within 30 min after LSD intake. These effects of LSD may persist for 12–18 h, even though the half-life of the drug is only 3 h.

The most frequent acute medical emergency associated with LSD use is a panic episode (the “bad trip”), which may persist up to 24 h. Management of this problem is best accomplished by supportive reassurance (“talking down”) and, if necessary, administration of small doses of anxiolytic drugs. Adverse consequences of chronic LSD use include enhanced risk for schizophreniform psychosis and derangements in memory function, problem solving, and abstract thinking. Treatment of these disorders is best carried out in specialized psychiatric facilities.

Tolerance develops rapidly for LSD-induced changes in psychological function when the drug is used one or more times per day for >4 days. Abrupt abstinence following continued use does not produce withdrawal signs or symptoms. There have been no clinical reports of death caused by the direct effects of LSD.

PHENCYCLIDINE (PCP)

Phencyclidine (PCP), a cyclohexylamine derivative, is widely used in veterinary medicine to briefly immobilize large animals and is sometimes described as a dissociative anesthetic. PCP binds to ionotropic *N*-methyl-D-aspartate

(NMDA) receptors in the nervous system, blocking ion current through these channels. PCP is easily synthesized; its abusers are primarily young people and polydrug users. It is used orally, by smoking, by snorting, or by IV injection. It is also used as an adulterant in THC, LSD, amphetamine, or cocaine. The most common street preparation, *angel dust*, is a white granular powder that contains 50–100% of the drug. Low doses (5 mg) produce agitation, excitement, impaired motor coordination, dysarthria, and analgesia. Physical signs of intoxication may include horizontal or vertical nystagmus, flushing, diaphoresis, and hyperacusis. Behavioral changes include distortions of body image, disorganization of thinking, and feelings of estrangement. Higher doses of PCP (5–10 mg) may produce profuse salivation, vomiting, myoclonus, fever, stupor, or coma. PCP doses of ≥10 mg cause convulsions, opisthotonus, and decerebrate posturing, which may be followed by prolonged coma.

The diagnosis of PCP overdose is difficult because the patient's initial symptoms (anxiety, paranoia, delusions, hallucinations) may suggest an acute schizophrenic reaction. Confirmation of PCP use is possible by determination of PCP levels in serum or urine. PCP assays are available at most toxicologic centers. PCP remains in urine for 1–5 days following high-dose intake.

PCP overdose requires life-support measures, including treatment of coma, convulsions, and respiratory depression in an intensive care unit. There is no specific antidote or antagonist for PCP. PCP excretion from the body can be enhanced by gastric lavage and acidification of urine. Death from PCP overdose may occur as a consequence of some combination of pharyngeal hypersecretion, hyperthermia, respiratory depression, severe hypertension, seizures, hypertensive encephalopathy, and intracerebral hemorrhage.

Acute psychosis associated with PCP use is a psychiatric emergency since patients may be at high risk for suicide or extreme violence toward others. Phenothiazines should not be used for treatment because these drugs potentiate PCP's anticholinergic effects. Haloperidol (5 mg IM) has been administered on an hourly basis to induce suppression of psychotic behavior. PCP, like LSD and mescaline, produces vasospasm of cerebral arteries at relatively low doses. Chronic PCP use has been shown to induce insomnia, anorexia, severe social and behavioral changes, and, in some cases, chronic schizophrenia.

OTHER DRUGS OF ABUSE

A number of other pharmacologically diverse drugs of abuse are often referred to as “club drugs” because these are frequently used in bars, at concerts, and rave parties. Commonly abused club drugs include flunitrazepam,

GHB, and ketamine and are described next. Methamphetamine, MDMA, and LSD are also considered club drugs and were described earlier in this chapter. Abuse of club drugs at high doses, especially in combination with alcohol, can be lethal and should be treated as a medical emergency. GHB and ketamine can be identified in blood, and flunitrazepam can be identified in urine and hair samples. Flunitrazepam and GHB toxicity can be treated with antagonists at benzodiazepine and GABA_B receptors, respectively.

Flunitrazepam

Flunitrazepam (Rohypnol) is a benzodiazepine derivative primarily used for treatment of insomnia, but it has significant abuse potential because of its strong hypnotic, anxiolytic, and amnesia-producing effects. It is a club drug commonly referred to as a “date-rape drug” or “roofies.” The drug enhances GABA_A receptor activity, and overdose can be treated with flumazenil, a benzodiazepine receptor antagonist. Flunitrazepam is typically used orally but can be snorted or injected. Concomitant use of alcohol or opiates is common, and this enhances the sedative and hypnotic effects of flunitrazepam and also the risk of motor vehicle accidents. Overdose can produce life-threatening respiratory depression and coma. Abrupt cessation after chronic use may result in a benzodiazepine withdrawal syndrome consisting of anxiety, insomnia, disordered thinking, and seizures.

GHB

Gamma-hydroxybutyric acid (Xyrem) is a sedative drug that is FDA-approved for the treatment of narcolepsy. It is classified as a club drug, is sometimes used in combination with alcohol or other drugs of abuse, and has been implicated in cases of date rape. GHB is usually taken orally and has no distinctive color or odor. Its stimulant properties are attributed to agonist activity at the GHB receptor, but it also has sedative effects at high doses that reflect its activity at GABA_B receptors. GABA_B antagonists can reverse GHB’s sedative effects, and opioid antagonists (naloxone, naltrexone) can attenuate GHB effects on dopamine release. Low doses of GHB may produce euphoria and disinhibition, whereas high doses result in nausea, agitation, convulsions, and sedation that can lead to unconsciousness and death from respiratory depression.

Ketamine

Ketamine (Ketaset, Ketalar) is a dissociative anesthetic, similar to phencyclidine (PCP). In veterinary medicine, it is used for brief immobilization. In clinical medicine, it is used for sedation, analgesia, and to

supplement anesthesia. Ketamine increases heart rate and blood pressure, with less respiratory depression than other anesthetics. Ketamine’s popularity as a club drug appears to reflect its ability to induce a dissociative state and feelings of depersonalization, accompanied by intense hallucinations and subsequent amnesia. It can be administered orally, by smoking (usually in combination with tobacco and/or marijuana), or by IV or IM injection. Like PCP, it binds to NMDA receptors and acts as an uncompetitive NMDA antagonist. Ketamine has a complex profile of action and appears to be useful as an antidepressant in treatment-resistant patients and as an analgesic in chronic-pain patients. The extent to which chronic recreational use leads to memory impairment remains controversial.

POLYDRUG ABUSE

Although some drug abusers may prefer a particular drug, the concurrent use of multiple drugs is often reported. Polydrug abuse often involves substances that may have different pharmacologic effects from the preferred drug. For example, concurrent use of such dissimilar compounds as stimulants and opiates or stimulants and alcohol is common. The diversity of reported drug use combinations suggests that achieving a subjective change in state, rather than any particular direction of change (stimulation or sedation), may be the primary reinforcer in polydrug abuse. There is also evidence that intoxication with alcohol, opiates, and cocaine is associated with increased tobacco smoking. There are relatively few controlled studies of multiple drug interactions. However, the combined use of cocaine, heroin, and alcohol increases the risk for toxic effects and adverse medical consequences. One determinant of polydrug use patterns is the relative availability and cost of the drugs. For example, alcohol abuse, with its attendant medical complications, is one of the most serious problems encountered in former heroin addicts participating in methadone maintenance programs.

The physician must recognize that perpetuation of polydrug abuse and drug dependence is not necessarily a symptom of an underlying emotional disorder. Neither alleviation of anxiety nor reduction of depression accounts for initiation and perpetuation of polydrug abuse. Severe depression and anxiety are the consequences of polydrug abuse as frequently as they are the antecedents. Interestingly, some adverse consequences of drug use may be reinforcing and contribute to the continuation of polydrug abuse.

Adequate treatment of polydrug abuse, as well as other forms of drug abuse, requires innovative intervention programs. The first step in successful treatment is detoxification, a process that may be difficult when

several drugs with different pharmacologic actions (e.g., alcohol, opiates, and cocaine) have been abused. Since patients may not recall or may deny simultaneous multiple drug use, diagnostic evaluation should always include urinalysis for qualitative detection of psychoactive substances and their metabolites. Treatment of polydrug abuse often requires hospitalization or inpatient residential care during detoxification and the initial phase of drug abstinence. When possible, specialized

facilities for the care and treatment of drug-dependent persons should be used. Outpatient detoxification of polydrug abuse patients is likely to be ineffective and may be dangerous.

Drug abuse disorders often respond to effective treatment, but episodes of relapse may occur unpredictably. The physician should continue to assist patients during relapse and recognize that occasional recurrent drug use is not unusual in this complex behavioral disorder.

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APPENDIX

LABORATORY VALUES OF CLINICAL IMPORTANCE



Alexander Kratz ■ Michael A. Pesce ■ Robert C. Basner
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This Appendix contains tables of reference values for laboratory tests, special analytes, and special function tests. A variety of factors can influence reference values. Such variables include the population studied, the duration and means of specimen transport, laboratory methods and instrumentation, and even the type of container used for the collection of the specimen. The reference or “normal” ranges given in this appendix may therefore not be appropriate for all laboratories, and these values should only be used as general guidelines. Whenever possible, reference values provided by the laboratory performing the testing should be utilized in the interpretation of laboratory data. Values supplied in this Appendix reflect typical reference ranges in adults. Pediatric reference ranges may vary significantly from adult values.

In preparing the Appendix, the authors have taken into account the fact that the system of international

units (SI, système international d’unités) is used in most countries and in some medical journals. However, clinical laboratories may continue to report values in “traditional” or conventional units. Therefore, both systems are provided in the Appendix. The dual system is also used in the text except for (1) those instances in which the numbers remain the same but only the terminology is changed (mmol/L for meq/L or IU/L for mIU/mL), when only the SI units are given; and (2) most pressure measurements (e.g., blood and cerebrospinal fluid pressures), when the traditional units (mmHg, mmH₂O) are used. In all other instances in the text the SI unit is followed by the traditional unit in parentheses.

REFERENCE VALUES FOR LABORATORY TESTS

TABLE 1

HEMATOLOGY AND COAGULATION

ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Activated clotting time	WB	70–180 s	70–180 s
Activated protein C resistance (factor V Leiden)	P	Not applicable	Ratio >2.1
ADAMTS13 activity	P	≥0.67	≥67%
ADAMTS13 inhibitor activity	P	Not applicable	≤0.4 U
ADAMTS13 antibody	P	Not applicable	≤18 U
Alpha ₂ antiplasmin	P	0.87–1.55	87–155%
Antiphospholipid antibody panel			
PTT-LA (lupus anticoagulant screen)	P	Negative	Negative
Platelet neutralization procedure	P	Negative	Negative
Dilute viper venom screen	P	Negative	Negative
Anticardiolipin antibody	S		
IgG		0–15 arbitrary units	0–15 GPL
IgM		0–15 arbitrary units	0–15 MPL

(continued)

TABLE 1
HEMATOLOGY AND COAGULATION (CONTINUED)

ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Antithrombin III	P		
Antigenic		220–390 mg/L	22–39 mg/dL
Functional		0.7–1.30 U/L	70–130%
Anti-Xa assay (heparin assay)	P		
Unfractionated heparin		0.3–0.7 kIU/L	0.3–0.7 IU/mL
Low-molecular-weight heparin		0.5–1.0 kIU/L	0.5–1.0 IU/mL
Danaparoid (Orgaran)		0.5–0.8 kIU/L	0.5–0.8 IU/mL
Autohemolysis test	WB	0.004–0.045	0.4–4.50%
Autohemolysis test with glucose	WB	0.003–0.007	0.3–0.7%
Bleeding time (adult)		<7.1 min	<7.1 min
Bone marrow: See Table 7			
Clot retraction	WB	0.50–1.00/2 h	50–100%/2 h
Cryofibrinogen	P	Negative	Negative
D-dimer	P	220–740 ng/mL FEU	220–740 ng/mL FEU
Differential blood count	WB		
Relative counts:			
Neutrophils		0.40–0.70	40–70%
Bands		0.0–0.05	0–5%
Lymphocytes		0.20–0.50	20–50%
Monocytes		0.04–0.08	4–8%
Eosinophils		0.0–0.6	0–6%
Basophils		0.0–0.02	0–2%
Absolute counts:			
Neutrophils		$1.42\text{--}6.34 \times 10^9/\text{L}$	1420–6340/mm ³
Bands		$0\text{--}0.45 \times 10^9/\text{L}$	0–450/mm ³
Lymphocytes		$0.71\text{--}4.53 \times 10^9/\text{L}$	710–4530/mm ³
Monocytes		$0.14\text{--}0.72 \times 10^9/\text{L}$	140–720/mm ³
Eosinophils		$0\text{--}0.54 \times 10^9/\text{L}$	0–540/mm ³
Basophils		$0\text{--}0.18 \times 10^9/\text{L}$	0–180/mm ³
Erythrocyte count	WB		
Adult males		$4.30\text{--}5.60 \times 10^{12}/\text{L}$	$4.30\text{--}5.60 \times 10^6/\text{mm}^3$
Adult females		$4.00\text{--}5.20 \times 10^{12}/\text{L}$	$4.00\text{--}5.20 \times 10^6/\text{mm}^3$
Erythrocyte life span	WB		
Normal survival		120 days	120 days
Chromium labeled, half-life ($t_{1/2}$)		25–35 days	25–35 days
Erythrocyte sedimentation rate	WB		
Females		0–20 mm/h	0–20 mm/h
Males		0–15 mm/h	0–15 mm/h
Euglobulin lysis time	P	7200–14,400 s	120–240 min
Factor II, prothrombin	P	0.50–1.50	50–150%
Factor V	P	0.50–1.50	50–150%
Factor VII	P	0.50–1.50	50–150%
Factor VIII	P	0.50–1.50	50–150%
Factor IX	P	0.50–1.50	50–150%
Factor X	P	0.50–1.50	50–150%
Factor XI	P	0.50–1.50	50–150%
Factor XII	P	0.50–1.50	50–150 %
Factor XIII screen	P	Not applicable	Present
Factor inhibitor assay	P	<0.5 Bethesda Units	<0.5 Bethesda Units
Fibrin(ogen) degradation products	P	0–1 mg/L	0–1 µg/mL
Fibrinogen	P	2.33–4.96 g/L	233–496 mg/dL
Glucose-6-phosphate dehydrogenase (erythrocyte)	WB	<2400 s	<40 min
Ham’s test (acid serum)	WB	Negative	Negative

(continued)

TABLE 1

HEMATOLOGY AND COAGULATION (CONTINUED)

ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Hematocrit	WB		
Adult males		0.388–0.464	38.8–46.4
Adult females		0.354–0.444	35.4–44.4
Hemoglobin			
Plasma	P	6–50 mg/L	0.6–5.0 mg/dL
Whole blood:	WB		
Adult males		133–162 g/L	13.3–16.2 g/dL
Adult females		120–158 g/L	12.0–15.8 g/dL
Hemoglobin electrophoresis	WB		
Hemoglobin A		0.95–0.98	95–98%
Hemoglobin A ₂		0.015–0.031	1.5–3.1%
Hemoglobin F		0–0.02	0–2.0%
Hemoglobins other than A, A ₂ , or F		Absent	Absent
Heparin-induced thrombocytopenia antibody	P	Negative	Negative
Immature platelet fraction (IPF)	WB	0.011–0.061	1.1–6.1%
Joint fluid crystal	JF	Not applicable	No crystals seen
Joint fluid mucin	JF	Not applicable	Only type I mucin present
Leukocytes			
Alkaline phosphatase (LAP)	WB	0.2–1.6 μ kat/L	13–100 μ /L
Count (WBC)	WB	$3.54\text{--}9.06 \times 10^9/\text{L}$	$3.54\text{--}9.06 \times 10^3/\text{mm}^3$
Mean corpuscular hemoglobin (MCH)	WB	26.7–31.9 pg/cell	26.7–31.9 pg/cell
Mean corpuscular hemoglobin concentration (MCHC)	WB	323–359 g/L	32.3–35.9 g/dL
Mean corpuscular hemoglobin of reticulocytes (CH)	WB	24–36 pg	24–36 pg
Mean corpuscular volume (MCV)	WB	79–93.3 fL	79–93.3 μm^3
Mean platelet volume (MPV)	WB	9.00–12.95 fL	9.00–12.95
Osmotic fragility of erythrocytes	WB		
Direct		0.0035–0.0045	0.35–0.45%
Indirect		0.0030–0.0065	0.30–0.65%
Partial thromboplastin time, activated	P	26.3–39.4 s	26.3–39.4 s
Plasminogen	P		
Antigen		84–140 mg/L	8.4–14.0 mg/dL
Functional		0.70–1.30	70–130%
Plasminogen activator inhibitor 1	P	4–43 $\mu\text{g}/\text{L}$	4–43 ng/mL
Platelet aggregation	PRP	Not applicable	>65% aggregation in response to adenosine diphosphate, epinephrine, collagen, ristocetin, and arachidonic acid
Platelet count	WB	$165\text{--}415 \times 10^9/\text{L}$	$165\text{--}415 \times 10^3/\text{mm}^3$
Platelet, mean volume	WB	6.4–11 fL	6.4–11.0 μm^3
Prekallikrein assay	P	0.50–1.5	50–150%
Prekallikrein screen	P		No deficiency detected
Protein C	P		
Total antigen		0.70–1.40	70–140%
Functional		0.70–1.30	70–130%
Protein S	P		
Total antigen		0.70–1.40	70–140%
Functional		0.65–1.40	65–140%
Free antigen		0.70–1.40	70–140%
Prothrombin gene mutation G20210A	WB	Not applicable	Not present
Prothrombin time	P	12.7–15.4 s	12.7–15.4 s

(continued)

TABLE 1

HEMATOLOGY AND COAGULATION (CONTINUED)			
ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Protoporphyrin, free erythrocyte	WB	0.28–0.64 μmol/L of red blood cells	16–36 μg/dL of red blood cells
Red cell distribution width	WB	<0.145	<14.5%
Reptilase time	P	16–23.6 s	16–23.6 s
Reticulocyte count	WB		
Adult males		0.008–0.023 red cells	0.8–2.3% red cells
Adult females		0.008–0.020 red cells	0.8–2.0% red cells
Reticulocyte hemoglobin content	WB	>26 pg/cell	>26 pg/cell
Ristocetin cofactor (functional von Willebrand factor)	P		
Blood group O		0.75 mean of normal	75% mean of normal
Blood group A		1.05 mean of normal	105% mean of normal
Blood group B		1.15 mean of normal	115% mean of normal
Blood group AB		1.25 mean of normal	125% mean of normal
Serotonin release assay	S	<0.2 release	<20% release
Sickle cell test	WB	Negative	Negative
Sucrose hemolysis	WB	<0.1	<10% hemolysis
Thrombin time	P	15.3–18.5 s	15.3–18.5 s
Total eosinophils	WB	150–300 × 10 ⁶ /L	150–300/mm ³
Transferrin receptor	S, P	9.6–29.6 nmol/L	9.6–29.6 nmol/L
Viscosity			
Plasma	P	1.7–2.1	1.7–2.1
Serum	S	1.4–1.8	1.4–1.8
von Willebrand factor (vWF) antigen (factor VIII:R antigen)			
Blood group O		0.75 mean of normal	75% mean of normal
Blood group A		1.05 mean of normal	105% mean of normal
Blood group B		1.15 mean of normal	115% mean of normal
Blood group AB		1.25 mean of normal	125% mean of normal
von Willebrand factor multimers	P	Normal distribution	Normal distribution
White blood cells: see “Leukocytes”			

Abbreviations: JF, joint fluid; P, plasma; PRP, platelet-rich plasma; S, serum; WB, whole blood.

TABLE 2

CLINICAL CHEMISTRY AND IMMUNOLOGY			
ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Acetoacetate	P	49–294 μmol/L	0.5–3.0 mg/dL
Adrenocorticotropin (ACTH)	P	1.3–16.7 pmol/L	6.0–76.0 pg/mL
Alanine aminotransferase (ALT, SGPT)	S	0.12–0.70 μkat/L	7–41 U/L
Albumin	S	40–50 g/L	4.0–5.0 mg/dL
Aldolase	S	26–138 nkat/L	1.5–8.1 U/L
Aldosterone (adult)			
Supine, normal sodium diet	S, P	<443 pmol/L	<16 ng/dL
Upright, normal	S, P	111–858 pmol/L	4–31 ng/dL
Alpha fetoprotein (adult)	S	0–8.5 μg/L	0–8.5 ng/mL
Alpha ₁ antitrypsin	S	1.0–2.0 g/L	100–200 mg/dL
Ammonia, as NH ₃	P	11–35 μmol/L	19–60 μg/dL
Amylase (method dependent)	S	0.34–1.6 μkat/L	20–96 U/L

(continued)

TABLE 2

CLINICAL CHEMISTRY AND IMMUNOLOGY (CONTINUED)

ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Androstendione (adult)	S		
Males		0.81–3.1 nmol/L	23–89 ng/dL
Females			
Premenopausal		0.91–7.5 nmol/L	26–214 ng/dL
Postmenopausal		0.46–2.9 nmol/L	13–82 ng/dL
Angiotensin-converting enzyme (ACE)	S	0.15–1.1 μ kat/L	9–67 U/L
Anion gap	S	7–16 mmol/L	7–16 mmol/L
Apolipoprotein A-1	S		
Male		0.94–1.78 g/L	94–178 mg/dL
Female		1.01–1.99 g/L	101–199 mg/dL
Apolipoprotein B	S		
Male		0.55–1.40 g/L	55–140 mg/dL
Female		0.55–1.25 g/L	55–125 mg/dL
Arterial blood gases	WB		
[HCO ₃ ⁻]		22–30 mmol/L	22–30 meq/L
PCO ₂		4.3–6.0 kPa	32–45 mmHg
pH		7.35–7.45	7.35–7.45
PO ₂		9.6–13.8 kPa	72–104 mmHg
Aspartate aminotransferase (AST, SGOT)	S	0.20–0.65 μ kat/L	12–38 U/L
Autoantibodies	S		
Anti-centromere antibody IgG		≤29 AU/mL	≤29 AU/mL
Anti-double-strand (native) DNA		<25 IU/L	<25 IU/L
Anti-glomerular basement membrane antibodies			
Qualitative IgG, IgA		Negative	Negative
Quantitative IgG antibody		≤19 AU/mL	≤19 AU/mL
Anti-histone antibodies		<1.0 U	<1.0 U
Anti-Jo-1 antibody		≤29 AU/mL	≤29 AU/mL
Anti-mitochondrial antibody		Not applicable	<20 Units
Anti-neutrophil cytoplasmic autoantibodies		Not applicable	<1:20
Serine proteinase 3 antibodies		≤19 AU/mL	≤19 AU/mL
Myeloperoxidase antibodies		≤19 AU/mL	≤19 AU/mL
Antinuclear antibody		Not applicable	Negative at 1:40
Anti-parietal cell antibody		Not applicable	None detected
Anti-RNP antibody		Not applicable	<1.0 U
Anti-Scl 70 antibody		Not applicable	<1.0 U
Anti-Smith antibody		Not applicable	<1.0 U
Anti-smooth muscle antibody		Not applicable	<1.0 U
Anti-SSA antibody		Not applicable	<1.0 U
Anti-SSB antibody		Not applicable	Negative
Anti-thyroglobulin antibody		<40 kIU/L	<40 IU/mL
Anti-thyroid peroxidase antibody		<35 kIU/L	<35 IU/mL
B-type natriuretic peptide (BNP)	P	Age and gender specific: <100 ng/L	Age and gender specific: <100 pg/mL
Bence Jones protein, serum qualitative	S	Not applicable	None detected
Bence Jones protein, serum quantitative	S		
Free kappa		3.3–19.4 mg/L	0.33–1.94 mg/dL
Free lambda		5.7–26.3 mg/L	0.57–2.63 mg/dL
K/L ratio		0.26–1.65	0.26–1.65
Beta-2-microglobulin	S	1.1–2.4 mg/L	1.1–2.4 mg/L
Bilirubin	S		
Total		5.1–22 μ mol/L	0.3–1.3 mg/dL
Direct		1.7–6.8 μ mol/L	0.1–0.4 mg/dL
Indirect		3.4–15.2 μ mol/L	0.2–0.9 mg/dL

(continued)

TABLE 2

CLINICAL CHEMISTRY AND IMMUNOLOGY (CONTINUED)			
ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
C peptide	S	0.27–1.19 nmol/L	0.8–3.5 ng/mL
C1-esterase-inhibitor protein	S	210–390 mg/L	21–39 mg/dL
CA 125	S	<35 kU/L	<35 U/mL
CA 19-9	S	<37 kU/L	<37 U/mL
CA 15-3	S	<33 kU/L	<33 U/mL
CA 27-29	S	0–40 kU/L	0–40 U/mL
Calcitonin	S		
Male		0–7.5 ng/L	0–7.5 pg/mL
Female		0–5.1 ng/L	0–5.1 pg/mL
Calcium	S	2.2–2.6 mmol/L	8.7–10.2 mg/dL
Calcium, ionized	WB	1.12–1.32 mmol/L	4.5–5.3 mg/dL
Carbon dioxide content (TCO ₂)	P (sea level)	22–30 mmol/L	22–30 meq/L
Carboxyhemoglobin (carbon monoxide content)	WB		
Nonsmokers		0.0–0.015	0–1.5%
Smokers		0.04–0.09	4–9%
Loss of consciousness and death		>0.50	>50%
Carcinoembryonic antigen (CEA)	S		
Nonsmokers		0.0–3.0 µg/L	0.0–3.0 ng/mL
Smokers		0.0–5.0 µg/L	0.0–5.0 ng/mL
Ceruloplasmin	S	250–630 mg/L	25–63 mg/dL
Chloride	S	102–109 mmol/L	102–109 meq/L
Cholesterol: see Table 5			
Cholinesterase	S	5–12 kU/L	5–12 U/mL
Chromogranin A	S	0–50 µg/L	0–50 ng/mL
Complement	S		
C3		0.83–1.77 g/L	83–177 mg/dL
C4		0.16–0.47 g/L	16–47 mg/dL
Complement total		60–144 CAE units	60–144 CAE units
Cortisol			
Fasting, 8 A.M.–12 noon	S	138–690 nmol/L	5–25 µg/dL
12 noon–8 P.M.		138–414 nmol/L	5–15 µg/dL
8 P.M.–8 A.M.		0–276 nmol/L	0–10 µg/dL
C-reactive protein	S	<10 mg/L	<10 mg/L
C-reactive protein, high sensitivity	S	Cardiac risk Low: <1.0 mg/L Average: 1.0–3.0 mg/L High: >3.0 mg/L	Cardiac risk Low: <1.0 mg/L Average: 1.0–3.0 mg/L High: >3.0 mg/L
Creatine kinase (total)	S		
Females		0.66–4.0 µkat/L	39–238 U/L
Males		0.875.0 µkat/L	51–294 U/L
Creatine kinase-MB	S		
Mass		0.0–5.5 µg/L	0.0–5.5 ng/mL
Fraction of total activity (by electrophoresis)		0–0.04	0–4.0%
Creatinine	S		
Female		44–80 µmol/L	0.5–0.9 mg/dL
Male		53–106 µmol/L	0.6–1.2 mg/dL
Cryoglobulins	S	Not applicable	None detected
Cystatin C	S	0.5–1.0 mg/L	0.5–1.0 mg/L

(continued)

TABLE 2

CLINICAL CHEMISTRY AND IMMUNOLOGY (CONTINUED)

ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Dehydroepiandrosterone (DHEA) (adult)			
Male	S	6.2–43.4 nmol/L	180–1250 ng/dL
Female		4.5–34.0 nmol/L	130–980 ng/dL
Dehydroepiandrosterone (DHEA) sulfate	S		
Male (adult)		100–6190 µg/L	10–619 µg/dL
Female (adult, premenopausal)		120–5350 µg/L	12–535 µg/dL
Female (adult, postmenopausal)		300–2600 µg/L	30–260 µg/dL
11-Deoxycortisol (adult) (compound S)	S	0.34–4.56 nmol/L	12–158 ng/dL
Dihydrotestosterone			
Male	S, P	1.03–2.92 nmol/L	30–85 ng/dL
Female		0.14–0.76 nmol/L	4–22 ng/dL
Dopamine	P	0–130 pmol/L	0–20 pg/mL
Epinephrine	P		
Supine (30 min)		<273 pmol/L	<50 pg/mL
Sitting		<328 pmol/L	<60 pg/mL
Standing (30 min)		<491 pmol/L	<90 pg/mL
Erythropoietin	S	4–27 U/L	4–27 U/L
Estradiol	S, P		
Female			
Menstruating:			
Follicular phase		74–532 pmol/L	<20–145 pg/mL
Midcycle peak		411–1626 pmol/L	112–443 pg/mL
Luteal phase		74–885 pmol/L	<20–241 pg/mL
Postmenopausal		217 pmol/L	<59 pg/mL
Male		74 pmol/L	<20 pg/mL
Estrone	S, P		
Female			
Menstruating:			
Follicular phase		<555 pmol/L	<150 pg/mL
Luteal phase		<740 pmol/L	<200 pg/mL
Postmenopausal		11–118 pmol/L	3–32 pg/mL
Male		33–133 pmol/L	9–36 pg/mL
Fatty acids, free (nonesterified)	P	0.1–0.6 mmol/L	2.8–16.8 mg/dL
Ferritin	S		
Female		10–150 µg/L	10–150 ng/mL
Male		29–248 µg/L	29–248 ng/mL
Follicle-stimulating hormone (FSH)	S, P		
Female			
Menstruating:			
Follicular phase		3.0–20.0 IU/L	3.0–20.0 mIU/mL
Ovulatory phase		9.0–26.0 IU/L	9.0–26.0 mIU/mL
Luteal phase		1.0–12.0 IU/L	1.0–12.0 mIU/mL
Postmenopausal		18.0–153.0 IU/L	18.0–153.0 mIU/mL
Male		1.0–12.0 IU/L	1.0–12.0 mIU/mL
Fructosamine	S	<285 µmol/L	<285 µmol/L
Gamma glutamyltransferase	S	0.15–0.99 µkat/L	9–58 U/L
Gastrin	S	<100 ng/L	<100 pg/mL
Glucagon	P	40–130 ng/L	40–130 pg/mL

(continued)

TABLE 2

CLINICAL CHEMISTRY AND IMMUNOLOGY (CONTINUED)			
ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Glucose	WB	3.6–5.3 mmol/L	65–95 mg/dL
Glucose (fasting)	P		
Normal		4.2–5.6 mmol/L	75–100 mg/dL
Increased risk for diabetes		5.6–6.9 mmol/L	100–125 mg/dL
Diabetes mellitus		Fasting >7.0 mmol/L	Fasting >126 mg/dL
		A 2-hour level of >11.1 mmol/L during an oral glucose tolerance test	A 2-hour level of ≥200 mg/dL during an oral glucose tolerance test
		A random glucose level of ≥11.1 mmol/L in patients with symptoms of hyperglycemia	A random glucose level of ≥200 mg/dL in patients with symptoms of hyperglycemia
Growth hormone	S	0–5 µg/L	0–5 ng/mL
Hemoglobin A _{1c}	WB	0.04–0.06 HgB fraction	4.0–5.6%
Pre-diabetes		0.057–0.064 HgB fraction	5.7–6.4%
Diabetes mellitus		A hemoglobin A _{1c} level of ≥0.065 Hgb fraction as suggested by the American Diabetes Association	A hemoglobin A _{1c} level of ≥6.5% as suggested by the American Diabetes Association
Hemoglobin A _{1c} with estimated average glucose (eAg)	WB	eAg mmol/L = 1.59 × HbA _{1c} – 2.59	eAg (mg/dL) = 28.7 × HbA _{1c} – 46.7
High-density lipoprotein (HDL) (see Table 5)			
Homocysteine	P	4.4–10.8 µmol/L	4.4–10.8 µmol/L
Human chorionic gonadotropin (HCG)	S		
Nonpregnant female		<5 IU/L	<5 mIU/mL
1–2 weeks postconception		9–130 IU/L	9–130 mIU/mL
2–3 weeks postconception		75–2600 IU/L	75–2600 mIU/mL
3–4 weeks postconception		850–20,800 IU/L	850–20,800 mIU/mL
4–5 weeks postconception		4000–100,200 IU/L	4000–100,200 mIU/mL
5–10 weeks postconception		11,500–289,000 IU/L	11,500–289,000 mIU/mL
10–14 weeks post conception		18,300–137,000 IU/L	18,300–137,000 mIU/mL
Second trimester		1400–53,000 IU/L	1400–53,000 mIU/mL
Third trimester		940–60,000 IU/L	940–60,000 mIU/mL
β-Hydroxybutyrate	P	60–170 µmol/L	0.6–1.8 mg/dL
17-Hydroxyprogesterone (adult)	S		
Male		<4.17 nmol/L	<139 ng/dL
Female			
Follicular phase		0.45–2.1 nmol/L	15–70 ng/dL
Luteal phase		1.05–8.7 nmol/L	35–290 ng/dL
Immunofixation	S	Not applicable	No bands detected
Immunoglobulin, quantitation (adult)			
IgA	S	0.70–3.50 g/L	70–350 mg/dL
IgD	S	0–140 mg/L	0–14 mg/dL
IgE	S	1–87 kIU/L	1–87 IU/mL
IgG	S	7.0–17.0 g/L	700–1700 mg/dL
IgG ₁	S	2.7–17.4 g/L	270–1740 mg/dL
IgG ₂	S	0.3–6.3 g/L	30–630 mg/dL
IgG ₃	S	0.13–3.2 g/L	13–320 mg/dL
IgG ₄	S	0.11–6.2 g/L	11–620 mg/dL
IgM	S	0.50–3.0 g/L	50–300 mg/dL
Insulin	S, P	14.35–143.5 pmol/L	2–20 µU/mL
Iron	S	7–25 µmol/L	41–141 µg/dL

(continued)

TABLE 2

CLINICAL CHEMISTRY AND IMMUNOLOGY (CONTINUED)

ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Iron-binding capacity	S	45–73 $\mu\text{mol/L}$	251–406 $\mu\text{g/dL}$
Iron-binding capacity saturation	S	0.16–0.35	16–35%
Ischemia modified albumin	S	<85 kU/L	<85 U/mL
Joint fluid crystal	JF	Not applicable	No crystals seen
Joint fluid mucin	JF	Not applicable	Only type I mucin present
Ketone (acetone)	S	Negative	Negative
Lactate	P, arterial P, venous	0.5–1.6 mmol/L 0.5–2.2 mmol/L	4.5–14.4 mg/dL 4.5–19.8 mg/dL
Lactate dehydrogenase	S	2.0–3.8 $\mu\text{kat/L}$	115–221 U/L
Lipase	S	0.51–0.73 $\mu\text{kat/L}$	3–43 U/L
Lipids: see Table 5			
Lipoprotein (a)	S	0–300 mg/L	0–30 mg/dL
Low-density lipoprotein (LDL) (see Table 5)			
Luteinizing hormone (LH)	S, P		
Female			
Menstruating:			
Follicular phase		2.0–15.0 U/L	2.0–15.0 mIU/mL
Ovulatory phase		22.0–105.0 U/L	22.0–105.0 mIU/mL
Luteal phase		0.6–19.0 U/L	0.6–19.0 mIU/mL
Postmenopausal		16.0–64.0 U/L	16.0–64.0 mIU/mL
Male		2.0–12.0 U/L	2.0–12.0 mIU/mL
Magnesium	S	0.62–0.95 mmol/L	1.5–2.3 mg/dL
Metanephrine	P	<0.5 nmol/L	<100 pg/mL
Methemoglobin	WB	0.0–0.01	0–1%
Myoglobin	S		
Male		20–71 $\mu\text{g/L}$	20–71 $\mu\text{g/L}$
Female		25–58 $\mu\text{g/L}$	25–58 $\mu\text{g/L}$
Norepinephrine	P		
Supine (30 min)		650–2423 pmol/L	110–410 pg/mL
Sitting		709–4019 pmol/L	120–680 pg/mL
Standing (30 min)		739–4137 pmol/L	125–700 pg/mL
N-telopeptide (cross-linked), NTx	S		
Female, premenopausal		6.2–19.0 nmol BCE	6.2–19.0 nmol BCE
Male		5.4–24.2 nmol BCE	5.4–24.2 nmol BCE
BCE = bone collagen equivalent			
NT-Pro BNP	S, P	<125 ng/L up to 75 years <450 ng/L >75 years	<125 pg/mL up to 75 years <450 pg/mL >75 years
5' Nucleotidase	S	0.00–0.19 $\mu\text{kat/L}$	0–11 U/L
Osmolality	P	275–295 mOsmol/kg serum water	275–295 mOsmol/kg serum water
Osteocalcin	S	11–50 $\mu\text{g/L}$	11–50 ng/mL
Oxygen content	WB		
Arterial (sea level)		17–21	17–21 vol%
Venous (sea level)		10–16	10–16 vol%
Oxygen saturation (sea level)	WB	Fraction:	Percent:
Arterial		0.94–1.0	94–100%
Venous, arm		0.60–0.85	60–85%
Parathyroid hormone (intact)	S	8–51 ng/L	8–51 pg/mL

(continued)

TABLE 2

CLINICAL CHEMISTRY AND IMMUNOLOGY (CONTINUED)			
ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
Phosphatase, alkaline	S	0.56–1.63 μ kat/L	33–96 U/L
Phosphorus, inorganic	S	0.81–1.4 mmol/L	2.5–4.3 mg/dL
Potassium	S	3.5–5.0 mmol/L	3.5–5.0 meq/L
Prealbumin	S	170–340 mg/L	17–34 mg/dL
Procalcitonin	S	<0.1 μ g/L	<0.1 ng/mL
Progesterone	S, P		
Female: Follicular		<3.18 nmol/L	<1.0 ng/mL
Midluteal		9.54–63.6 nmol/L	3–20 ng/mL
Male		<3.18 nmol/L	<1.0 ng/mL
Prolactin	S		
Male		53–360 mg/L	2.5–17 ng/mL
Female		40–530 mg/L	1.9–25 ng/mL
Prostate-specific antigen (PSA)	S	0.0–4.0 μ g/L	0.0–4.0 ng/mL
Prostate-specific antigen, free	S	With total PSA between 4 and 10 μ g/L and when the free PSA is: >0.25 decreased risk of prostate cancer <0.10 increased risk of prostate cancer	With total PSA between 4 and 10 ng/mL and when the free PSA is: >25% decreased risk of prostate cancer <10% increased risk of prostate cancer
Protein fractions:	S		
Albumin		35–55 g/L	3.5–5.5 g/dL (50–60%)
Globulin		20–35 g/L	2.0–3.5 g/dL (40–50%)
Alpha ₁		2–4 g/L	0.2–0.4 g/dL (4.2–7.2%)
Alpha ₂		5–9 g/L	0.5–0.9 g/dL (6.8–12%)
Beta		6–11 g/L	0.6–1.1 g/dL (9.3–15%)
Gamma		7–17 g/L	0.7–1.7 g/dL (13–23%)
Protein, total	S	67–86 g/L	6.7–8.6 g/dL
Pyruvate	P	40–130 μ mol/L	0.35–1.14 mg/dL
Rheumatoid factor	S	<15 kIU/L	<15 IU/mL
Serotonin	WB	0.28–1.14 μ mol/L	50–200 ng/mL
Serum protein electrophoresis	S	Not applicable	Normal pattern
Sex hormone-binding globulin (adult)	S		
Male		11–80 nmol/L	11–80 nmol/L
Female		30–135 nmol/L	30–135 nmol/L
Sodium	S	136–146 mmol/L	136–146 meq/L
Somatomedin-C (IGF-1) (adult)	S		
16 years		226–903 μ g/L	226–903 ng/mL
17 years		193–731 μ g/L	193–731 ng/mL
18 years		163–584 μ g/L	163–584 ng/mL
19 years		141–483 μ g/L	141–483 ng/mL
20 years		127–424 μ g/L	127–424 ng/mL
21–25 years		116–358 μ g/L	116–358 ng/mL
26–30 years		117–329 μ g/L	117–329 ng/mL
31–35 years		115–307 μ g/L	115–307 ng/mL
36–40 years		119–204 μ g/L	119–204 ng/mL
41–45 years		101–267 μ g/L	101–267 ng/mL
46–50 years		94–252 μ g/L	94–252 ng/mL
51–55 years		87–238 μ g/L	87–238 ng/mL
56–60 years		81–225 μ g/L	81–225 ng/mL
61–65 years		75–212 μ g/L	75–212 ng/mL

(continued)

TABLE 2

CLINICAL CHEMISTRY AND IMMUNOLOGY (CONTINUED)

ANALYTE	SPECIMEN	SI UNITS	CONVENTIONAL UNITS
66–70 years		69–200 µg/L	69–200 ng/mL
71–75 years		64–188 µg/L	64–188 ng/mL
76–80 years		59–177 µg/L	59–177 ng/mL
81–85 years		55–166 µg/L	55–166 ng/mL
Somatostatin	P	<25 ng/L	<25 pg/mL
Testosterone, free			
Female, adult	S	10.4–65.9 pmol/L	3–19 pg/mL
Male, adult		312–1041 pmol/L	90–300 pg/mL
Testosterone, total,	S		
Female		0.21–2.98 nmol/L	6–86 ng/dL
Male		9.36–37.10 nmol/L	270–1070 ng/dL
Thyroglobulin	S	1.3–31.8 µg/L	1.3–31.8 ng/mL
Thyroid-binding globulin	S	13–30 mg/L	1.3–3.0 mg/dL
Thyroid-stimulating hormone	S	0.34–4.25 mIU/L	0.34–4.25 µIU/mL
Thyroxine, free (fT4)	S	9.0–16 pmol/L	0.7–1.24 ng/dL
Thyroxine, total (T4)	S	70–151 nmol/L	5.4–11.7 µg/dL
Thyroxine index (free)	S	6.7–10.9	6.7–10.9
Transferrin	S	2.0–4.0 g/L	200–400 mg/dL
Triglycerides (see Table 5)	S	0.34–2.26 mmol/L	30–200 mg/dL
Triiodothyronine, free (fT3)	S	3.7–6.5 pmol/L	2.4–4.2 pg/mL
Triiodothyronine, total (T3)	S	1.2–2.1 nmol/L	77–135 ng/dL
Troponin I (method dependent)	S, P		
99th percentile of a healthy population		0–0.04 µg/L	0–0.04 ng/mL
Troponin T	S, P		
99th percentile of a healthy population		0–0.01 µg/L	0–0.01 ng/mL
Urea nitrogen	S	2.5–7.1 mmol/L	7–20 mg/dL
Uric acid	S		
Females		0.15–0.33 mmol/L	2.5–5.6 mg/dL
Males		0.18–0.41 mmol/L	3.1–7.0 mg/dL
Vasoactive intestinal polypeptide	P	0–60 ng/L	0–60 pg/mL
Zinc protoporphyrin	WB	0–400 µg/L	0–40 µg/dL
Zinc protoporphyrin (ZPP)-to-heme ratio	WB	0–69 µmol ZPP/mol heme	0–69 µmol ZPP/mol heme

Abbreviations: P, plasma; S, serum; WB, whole blood.

TABLE 3

TOXICOLOGY AND THERAPEUTIC DRUG MONITORING

DRUG	THERAPEUTIC RANGE		TOXIC LEVEL	
	SI UNITS	CONVENTIONAL UNITS	SI UNITS	CONVENTIONAL UNITS
Acetaminophen	66–199 µmol/L	10–30 µg/mL	>1320 µmol/L	>200 µg/mL
Amikacin				
Peak	34–51 µmol/L	20–30 µg/mL	>60 µmol/L	>35 µg/mL
Trough	0–17 µmol/L	0–10 µg/mL	>17 µmol/L	>10 µg/mL
Amitriptyline/nortriptyline (total drug)	430–900 nmol/L	120–250 ng/mL	>1800 nmol/L	>500 ng/mL
Amphetamine	150–220 nmol/L	20–30 ng/mL	>1500 nmol/L	>200 ng/mL
Bromide	9.4–18.7 mmol/L	75–150 mg/dL	>18.8 mmol/L	>150 mg/dL
Mild toxicity			6.4–18.8 mmol/L	51–150 mg/dL
Severe toxicity			>18.8 mmol/L	>150 mg/dL
Lethal			>37.5 mmol/L	>300 mg/dL
Caffeine	25.8–103 µmol/L	5–20 µg/mL	>206 µmol/L	>40 µg/mL
Carbamazepine	17–42 µmol/L	4–10 µg/mL	>85 µmol/L	>20 µg/mL
Chloramphenicol				
Peak	31–62 µmol/L	10–20 µg/mL	>77 µmol/L	>25 µg/mL
Trough	15–31 µmol/L	5–10 µg/mL	>46 µmol/L	>15 µg/mL
Chlordiazepoxide	1.7–10 µmol/L	0.5–3.0 µg/mL	>17 µmol/L	>5.0 µg/mL
Clonazepam	32–240 nmol/L	10–75 ng/mL	>320 nmol/L	>100 ng/mL
Clozapine	0.6–2.1 µmol/L	200–700 ng/mL	>3.7 µmol/L	>1200 ng/mL
Cocaine			>3.3 µmol/L	>1.0 µg/mL
Codeine	43–110 nmol/mL	13–33 ng/mL	>3700 nmol/mL	>1100 ng/mL (lethal)
Cyclosporine				
Renal transplant				
0–6 months	208–312 nmol/L	250–375 ng/mL	>312 nmol/L	>375 ng/mL
6–12 months after transplant	166–250 nmol/L	200–300 ng/mL	>250 nmol/L	>300 ng/mL
>12 months	83–125 nmol/L	100–150 ng/mL	>125 nmol/L	>150 ng/mL
Cardiac transplant				
0–6 months	208–291 nmol/L	250–350 ng/mL	>291 nmol/L	>350 ng/mL
6–12 months after transplant	125–208 nmol/L	150–250 ng/mL	>208 nmol/L	>250 ng/mL
>12 months	83–125 nmol/L	100–150 ng/mL	>125 nmol/L	150 ng/mL
Lung transplant				
0–6 months	250–374 nmol/L	300–450 ng/mL	>374 nmol/L	>450 ng/mL
Liver transplant				
Initiation	208–291 nmol/L	250–350 ng/mL	>291 nmol/L	>350 ng/mL
Maintenance	83–166 nmol/L	100–200 ng/mL	>166 nmol/L	>200 ng/mL
Desipramine	375–1130 nmol/L	100–300 ng/mL	>1880 nmol/L	>500 ng/mL
Diazepam (and metabolite)				
Diazepam	0.7–3.5 µmol/L	0.2–1.0 µg/mL	>7.0 µmol/L	>2.0 µg/mL
Nordiazepam	0.4–6.6 µmol/L	0.1–1.8 µg/mL	>9.2 µmol/L	>2.5 µg/mL
Digoxin	0.64–2.6 nmol/L	0.5–2.0 ng/mL	>5.0 nmol/L	>3.9 ng/mL
Disopyramide	5.3–14.7 µmol/L	2–5 µg/mL	>20.6 µmol/L	>7 µg/mL
Doxepin and nordoxepin				
Doxepin	0.36–0.98 µmol/L	101–274 ng/mL	>1.8 µmol/L	>503 ng/mL
Nordoxepin	0.38–1.04 µmol/L	106–291 ng/mL	>1.9 µmol/L	>531 ng/mL
Ethanol				
Behavioral changes			>4.3 mmol/L	>20 mg/dL
Legal limit			≥17 mmol/L	≥80 mg/dL
Critical with acute exposure			>54 mmol/L	>250 mg/dL
Ethylene glycol				
Toxic			>2 mmol/L	>12 mg/dL
Lethal			>20 mmol/L	>120 mg/dL

(continued)

TABLE 3

TOXICOLOGY AND THERAPEUTIC DRUG MONITORING (CONTINUED)

DRUG	THERAPEUTIC RANGE		TOXIC LEVEL	
	SI UNITS	CONVENTIONAL UNITS	SI UNITS	CONVENTIONAL UNITS
Ethosuximide	280–700 µmol/L	40–100 µg/mL	>700 µmol/L	>100 µg/mL
Everolimus	3.13–8.35 nmol/L	3–8 ng/mL	>12.5 nmol/L	>12 ng/mL
Flecainide	0.5–2.4 µmol/L	0.2–1.0 µg/mL	>3.6 µmol/L	>1.5 µg/mL
Gentamicin				
Peak	10–21 µmol/mL	5–10 µg/mL	>25 µmol/mL	>12 µg/mL
Trough	0–4.2 µmol/mL	0–2 µg/mL	>4.2 µmol/mL	>2 µg/mL
Heroin (diacetyl morphine)			>700 µmol/L	>200 ng/mL (as morphine)
Ibuprofen	49–243 µmol/L	10–50 µg/mL	>970 µmol/L	>200 µg/mL
Imipramine (and metabolite)				
Desimipramine	375–1130 nmol/L	100–300 ng/mL	>1880 nmol/L	>500 ng/mL
Total imipramine + desimipramine	563–1130 nmol/L	150–300 ng/mL	>1880 nmol/L	>500 ng/mL
Lamotrigine	11.7–54.7 µmol/L	3–14 µg/mL	>58.7 µmol/L	>15 µg/mL
Lidocaine	5.1–21.3 µmol/L	1.2–5.0 µg/mL	>38.4 µmol/L	>9.0 µg/mL
Lithium	0.5–1.3 mmol/L	0.5–1.3 meq/L	>2 mmol/L	>2 meq/L
Methadone	1.0–3.2 µmol/L	0.3–1.0 µg/mL	>6.5 µmol/L	>2 µg/mL
Methamphetamine	0.07–0.34 µmol/L	0.01–0.05 µg/mL	>3.35 µmol/L	>0.5 µg/mL
Methanol			>6 mmol/L	>20 mg/dL
Methotrexate				
Low-dose	0.01–0.1 µmol/L	0.01–0.1 µmol/L	>0.1 mmol/L	>0.1 mmol/L
High-dose (24 h)	<5.0 µmol/L	<5.0 µmol/L	>5.0 µmol/L	>5.0 µmol/L
High-dose (48 h)	<0.50 µmol/L	<0.50 µmol/L	>0.5 µmol/L	>0.5 µmol/L
High-dose (72 h)	<0.10 µmol/L	<0.10 µmol/L	>0.1 µmol/L	>0.1 µmol/L
Morphine	232–286 µmol/L	65–80 ng/mL	>720 µmol/L	>200 ng/mL
Mycophenolic acid	3.1–10.9 µmol/L	1.0–3.5 ng/mL	>37 µmol/L	>12 ng/mL
Nitroprusside (as thiocyanate)	103–499 µmol/L	6–29 µg/mL	860 µmol/L	>50 µg/mL
Nortriptyline	190–569 nmol/L	50–150 ng/mL	>1900 nmol/L	>500 ng/mL
Phenobarbital	65–172 µmol/L	15–40 µg/mL	>258 µmol/L	>60 µg/mL
Phenytoin	40–79 µmol/L	10–20 µg/mL	>158 µmol/L	>40 µg/mL
Phenytoin, free	4.0–7.9 µg/mL	1–2 µg/mL	>13.9 µg/mL	>3.5 µg/mL
% Free	0.08–0.14	8–14%		
Primidone and metabolite				
Primidone	23–55 µmol/L	5–12 µg/mL	>69 µmol/L	>15 µg/mL
Phenobarbital	65–172 µmol/L	15–40 µg/mL	>215 µmol/L	>50 µg/mL
Procainamide				
Procainamide	17–42 µmol/L	4–10 µg/mL	>43 µmol/L	>10 µg/mL
NAPA (N-acetylprocainamide)	22–72 µmol/L	6–20 µg/mL	>126 µmol/L	>35 µg/mL
Quinidine	6.2–15.4 µmol/L	2.0–5.0 µg/mL	>19 µmol/L	>6 µg/mL
Salicylates	145–2100 µmol/L	2–29 mg/dL	>2900 µmol/L	>40 mg/dL
Sirolimus (trough level)				
Kidney transplant	4.4–15.4 nmol/L	4–14 ng/mL	>16 nmol/L	>15 ng/mL
Tacrolimus (FK506) (trough)				
Kidney and liver				
Initiation	12–19 nmol/L	10–15 ng/mL	>25 nmol/L	>20 ng/mL
Maintenance	6–12 nmol/L	5–10 ng/mL	>25 nmol/L	>20 ng/mL
Heart				
Initiation	19–25 nmol/L	15–20 ng/mL		
Maintenance	6–12 nmol/L	5–10 ng/mL		

(continued)

TABLE 3
TOXICOLOGY AND THERAPEUTIC DRUG MONITORING (CONTINUED)

DRUG	THERAPEUTIC RANGE		TOXIC LEVEL	
	SI UNITS	CONVENTIONAL UNITS	SI UNITS	CONVENTIONAL UNITS
Theophylline	56–111 µg/mL	10–20 µg/mL	>168 µg/mL	>30 µg/mL
Thiocyanate				
After nitroprusside infusion	103–499 µmol/L	6–29 µg/mL	860 µmol/L	>50 µg/mL
Nonsmoker	17–69 µmol/L	1–4 µg/mL		
Smoker	52–206 µmol/L	3–12 µg/mL		
Tobramycin				
Peak	11–22 µg/L	5–10 µg/mL	>26 µg/L	>12 µg/mL
Trough	0–4.3 µg/L	0–2 µg/mL	>4.3 µg/L	>2 µg/mL
Valproic acid	346–693 µmol/L	50–100 µg/mL	>693 µmol/L	>100 µg/mL
Vancomycin				
Peak	14–28 µmol/L	20–40 µg/mL	>55 µmol/L	>80 µg/mL
Trough	3.5–10.4 µmol/L	5–15 µg/mL	>14 µmol/L	>20 µg/mL

TABLE 4
VITAMINS AND SELECTED TRACE MINERALS

SPECIMEN	ANALYTE	REFERENCE RANGE	
		SI UNITS	CONVENTIONAL UNITS
Aluminum	S	<0.2 µmol/L	<5.41 µg/L
Arsenic	WB	0.03–0.31 µmol/L	2–23 µg/L
Cadmium	WB	<44.5 nmol/L	<5.0 µg/L
Coenzyme Q10 (ubiquinone)	P	433–1532 µg/L	433–1532 µg/L
β-Carotene	S	0.07–1.43 µmol/L	4–77 µg/dL
Copper	S	11–22 µmol/L	70–140 µg/dL
Folic acid	RC	340–1020 nmol/L cells	150–450 ng/mL cells
Folic acid	S	12.2–40.8 nmol/L	5.4–18.0 ng/mL
Lead (adult)	S	<0.5 µmol/L	<10 µg/dL
Mercury	WB	3.0–294 nmol/L	0.6–59 µg/L
Selenium	S	0.8–2.0 µmol/L	63–160 µg/L
Vitamin A	S	0.7–3.5 µmol/L	20–100 µg/dL
Vitamin B ₁ (thiamine)	S	0–75 nmol/L	0–2 µg/dL
Vitamin B ₂ (riboflavin)	S	106–638 nmol/L	4–24 µg/dL
Vitamin B ₆	P	20–121 nmol/L	5–30 ng/mL
Vitamin B ₁₂	S	206–735 pmol/L	279–996 pg/mL
Vitamin C (ascorbic acid)	S	23–57 µmol/L	0.4–1.0 mg/dL
Vitamin D ₃ , 1,25-dihydroxy, total	S, P	36–180 pmol/L	15–75 pg/mL
Vitamin D ₃ , 25-hydroxy, total	P	75–250 nmol/L	30–100 ng/mL
Vitamin E	S	12–42 µmol/L	5–18 µg/mL
Vitamin K	S	0.29–2.64 nmol/L	0.13–1.19 ng/mL
Zinc	S	11.5–18.4 µmol/L	75–120 µg/dL

Abbreviations: P, plasma; RC, red cells; S, serum; WB, whole blood.

TABLE 5

CLASSIFICATION OF LDL, TOTAL, AND HDL CHOLESTEROL

LDL Cholesterol	
<70 mg/dL	Therapeutic option for very high risk patients
<100 mg/dL	Optimal
100–129 mg/dL	Near optimal/above optimal
130–159 mg/dL	Borderline high
160–189 mg/dL	High
≥190 mg/dL	Very high
Total Cholesterol	
<200 mg/dL	Desirable
200–239 mg/dL	Borderline high
≥240 mg/dL	High
HDL Cholesterol	
<40 mg/dL	Low
≥60 mg/dL	High

Abbreviations: LDL, low-density lipoprotein; HDL, high-density lipoprotein.

Source: Executive summary of the third report of the National Cholesterol Education Program (NCEP) expert panel on detection, evaluation, and treatment of high blood cholesterol in adults (adult treatment panel III). JAMA 2001; 285:2486–97. Implications of Recent Clinical Trials for the National Cholesterol Education Program Adult Treatment Panel III Guidelines. SM Grundy et al for the Coordinating Committee of the National Cholesterol Education Program: Circulation 110:227, 2004.

REFERENCE VALUES FOR SPECIFIC ANALYTES

TABLE 6

CEREBROSPINAL FLUID^a

CONSTITUENT	REFERENCE RANGE	
	SI UNITS	CONVENTIONAL UNITS
Osmolarity	292–297 mmol/kg water	292–297 mOsm/L
Electrolytes		
Sodium	137–145 mmol/L	137–145 meq/L
Potassium	2.7–3.9 mmol/L	2.7–3.9 meq/L
Calcium	1.0–1.5 mmol/L	2.1–3.0 meq/L
Magnesium	1.0–1.2 mmol/L	2.0–2.5 meq/L
Chloride	116–122 mmol/L	116–122 meq/L
CO ₂ content	20–24 mmol/L	20–24 meq/L
Pco ₂	6–7 kPa	45–49 mmHg
pH	7.31–7.34	
Glucose	2.22–3.89 mmol/L	40–70 mg/dL
Lactate	1–2 mmol/L	10–20 mg/dL
Total protein:		
Lumbar	0.15–0.5 g/L	15–50 mg/dL
Cisternal	0.15–0.25 g/L	15–25 mg/dL
Ventricular	0.06–0.15 g/L	6–15 mg/dL
Albumin	0.066–0.442 g/L	6.6–44.2 mg/dL
IgG	0.009–0.057 g/L	0.9–5.7 mg/dL
IgG index ^b	0.29–0.59	
Oligoclonal bands (OGB)	<2 bands not present in matched serum sample	
Ammonia	15–47 μmol/L	25–80 μg/dL
Creatinine	44–168 μmol/L	0.5–1.9 mg/dL
Myelin basic protein	<4 μg/L	
CSF pressure		50–180 mmH ₂ O
CSF volume (adult)	~150 mL	
Red blood cells	0	0
Leukocytes		
Total	0–5 mononuclear cells per μL	
Differential		
Lymphocytes	60–70%	
Monocytes	30–50%	
Neutrophils	None	

^aSince cerebrospinal fluid concentrations are equilibrium values, measurements of the same parameters in blood plasma obtained at the same time are recommended. However, there is a time lag in attainment of equilibrium, and cerebrospinal levels of plasma constituents that can fluctuate rapidly (such as plasma glucose) may not achieve stable values until after a significant lag phase.

^bIgG index = CSF IgG (mg/dL) × serum albumin (g/dL)/serum IgG (g/dL) × CSF albumin (mg/dL).

TABLE 7A
DIFFERENTIAL NUCLEATED CELL COUNTS OF BONE MARROW ASPIRATES^a

	OBSERVED RANGE (%)	95% RANGE (%)	MEAN (%)
Blast cells	0–3.2	0–3.0	1.4
Promyelocytes	3.6–13.2	3.2–12.4	7.8
Neutrophil myelocytes	4–21.4	3.7–10.0	7.6
Eosinophil myelocytes	0–5.0	0–2.8	1.3
Metamyelocytes	1–7.0	2.3–5.9	4.1
Neutrophils			
Males	21.0–45.6	21.9–42.3	32.1
Females	29.6–46.6	28.8–45.9	37.4
Eosinophils	0.4–4.2	0.3–4.2	2.2
Eosinophils plus eosinophil myelocytes	0.9–7.4	0.7–6.3	3.5
Basophils	0–0.8	0–0.4	0.1
Erythroblasts			
Male	18.0–39.4	16.2–40.1	28.1
Females	14.0–31.8	13.0–32.0	22.5
Lymphocytes	4.6–22.6	6.0–20.0	13.1
Plasma cells	0–1.4	0–1.2	0.6
Monocytes	0–3.2	0–2.6	1.3
Macrophages	0–1.8	0–1.3	0.4
M:E ratio			
Males	1.1–4.0	1.1–4.1	2.1
Females	1.6–5.4	1.6–5.2	2.8

^aBased on bone marrow aspirate from 50 healthy volunteers (30 men, 20 women).
Abbreviation: M:E, myeloid to erythroid ratio.
Source: BJ Bain: Br J Haematol 94:206, 1996.

TABLE 7B
BONE MARROW CELLULARITY

AGE	OBSERVED RANGE	95% RANGE	MEAN
Under 10 years	59.0–95.1%	72.9–84.7%	78.8%
10–19 years	41.5–86.6%	59.2–69.4%	64.3%
20–29 years	32.0–83.7%	54.1–61.9%	58.0%
30–39 years	30.3–81.3%	41.1–54.1%	47.6%
40–49 years	16.3–75.1%	43.5–52.9%	48.2%
50–59 years	19.7–73.6%	41.2–51.4%	46.3%
60–69 years	16.3–65.7%	40.8–50.6%	45.7%
70–79 years	11.3–47.1%	22.6–35.2%	28.9%

Source: From RJ Hartsock et al: Am J Clin Pathol 1965; 43:326, 1965.

TABLE 8
STOOL ANALYSIS

	REFERENCE RANGE	
	SI UNITS	CONVENTIONAL UNITS
Alpha-1-antitrypsin	≤540 mg/L	≤54 mg/dL
Amount	0.1–0.2 kg/d	100–200 g/24 h
Coproporphyrin	611–1832 nmol/d	400–1200 μg/24 h
Fat		
Adult		<7 g/d
Adult on fat-free diet		<4 g/d
Fatty acids	0–21 mmol/d	0–6 g/24 h
Leukocytes	None	None
Nitrogen	<178 mmol/d	<2.5 g/24 h
pH	7.0–7.5	
Potassium	14–102 mmol/L	14–102 mmol/L
Occult blood	Negative	Negative
Osmolality	280–325 mOsmol/kg	280–325 mOsmol/kg
Sodium	7–72 mmol/L	7–72 mmol/L
Trypsin		20–95 U/g
Urobilinogen	85–510 μmol/d	50–300 mg/24 h
Uroporphyrins	12–48 nmol/d	10–40 μg/24 h
Water	<0.75	<75%

Source: Modified from: FT Fishbach, MB Dunning III: *A Manual of Laboratory and Diagnostic Tests*, 7th ed. Philadelphia, Lippincott Williams & Wilkins, 2004.

TABLE 9

URINE ANALYSIS AND RENAL FUNCTION TESTS

	REFERENCE RANGE	
	SI UNITS	CONVENTIONAL UNITS
Acidity, titratable	20–40 mmol/d	20–40 meq/d
Aldosterone	Normal diet: 6–25 µg/d Low-salt diet: 17–44 µg/d High-salt diet: 0–6 µg/d	Normal diet: 6–25 µg/d Low-salt diet: 17–44 µg/d High-salt diet: 0–6 µg/d
Aluminum	0.19–1.11 µmol/L	5–30 µg/L
Ammonia	30–50 mmol/d	30–50 meq/d
Amylase		4–400 U/L
Amylase/creatinine clearance ratio [(Cl _{am} /Cl _{cr}) × 100]	1–5	1–5
Arsenic	0.07–0.67 µmol/d	5–50 µg/d
Bence Jones protein, urine, qualitative	Not applicable	None detected
Bence Jones protein, urine, quantitative		
Free Kappa	1.4–24.2 mg/L	0.14–2.42 mg/dL
Free Lambda	0.2–6.7 mg/L	0.02–0.67 mg/dL
K/L ratio	2.04–10.37	2.04–10.37
Calcium (10 meq/d or 200 mg/d dietary calcium)	<7.5 mmol/d	<300 mg/d
Chloride	140–250 mmol/d	140–250 mmol/d
Citrate	320–1240 mg/d	320–1240 mg/d
Copper	<0.95 µmol/d	<60 µg/d
Coproporphyrins (types I and III)	0–20 µmol/mol creatinine	0–20 µmol/mol creatinine
Cortisol, free	55–193 nmol/d	20–70 µg/d
Creatine, as creatinine		
Female	<760 µmol/d	<100 mg/d
Male	<380 µmol/d	<50 mg/d
Creatinine	8.8–14 mmol/d	1.0–1.6 g/d
Dopamine	392–2876 nmol/d	60–440 µg/d
Eosinophils	<100 eosinophils/mL	<100 eosinophils/mL
Epinephrine	0–109 nmol/d	0–20 µg/d
Glomerular filtration rate	>60 mL/min/1.73 m ² For African Americans multiply the result by 1.21	>60 mL/min/1.73 m ² For African Americans multiply the result by 1.21
Glucose (glucose oxidase method)	0.3–1.7 mmol/d	50–300 mg/d
5-Hydroindoleacetic acid [5-HIAA]	0–78.8 µmol/d	0–15 mg/d
Hydroxyproline	53–328 µmol/d	53–328 µmol/d
Iodine, spot urine		
WHO classification of iodine deficiency:		
Not iodine deficient	>100 µg/L	>100 µg/L
Mild iodine deficiency	50–100 µg/L	50–100 µg/L
Moderate iodine deficiency	20–49 µg/L	20–49 µg/L
Severe iodine deficiency	<20 µg/L	<20 µg/L
Ketone (acetone)	Negative	Negative
17 Ketosteroids	3–12 mg/d	3–12 mg/d
Metanephrines		
Metanephrine	30–350 µg/d	30–350 µg/d
Normetanephrine	50–650 µg/d	50–650 µg/d

(continued)

TABLE 9
URINE ANALYSIS AND RENAL FUNCTION TESTS (CONTINUED)

	REFERENCE RANGE	
	SI UNITS	CONVENTIONAL UNITS
Microalbumin		
Normal	0.0–0.03 g/d	0–30 mg/d
Microalbuminuria	0.03–0.30 g/d	30–300 mg/d
Clinical albuminuria	>0.3 g/d	>300 mg/d
Microalbumin/creatinine ratio		
Normal	0–3.4 g/mol creatinine	0–30 µg/mg creatinine
Microalbuminuria	3.4–34 g/mol creatinine	30–300 µg/mg creatinine
Clinical albuminuria	>34 g/mol creatinine	>300 µg/mg creatinine
β ₂ -Microglobulin	0–160 µg/L	0–160 µg/L
Norepinephrine	89–473 nmol/d	15–80 µg/d
N-telopeptide (cross-linked), NTx		
Female, premenopausal	17–94 nmol BCE/mmol creatinine	17–94 nmol BCE/mmol creatinine
Female, postmenopausal	26–124 nmol BCE/mmol creatinine	26–124 nmol BCE/mmol creatinine
Male	21–83 nmol BCE/mmol creatinine	21–83 nmol BCE/mmol creatinine
BCE = bone collagen equivalent		
Osmolality	100–800 mOsm/kg	100–800 mOsm/kg
Oxalate		
Male	80–500 µmol/d	7–44 mg/d
Female	45–350 µmol/d	4–31 mg/d
pH	5.0–9.0	5.0–9.0
Phosphate (phosphorus) (varies with intake)	12.9–42.0 mmol/d	400–1300 mg/d
Porphobilinogen	None	None
Potassium (varies with intake)	25–100 mmol/d	25–100 meq/d
Protein	<0.15 g/d	<150 mg/d
Protein/creatinine ratio	Male: 15–68 mg/g Female: 10–107 mg/g	Male: 15–68 mg/g Female: 10–107 mg/g
Sediment		
Red blood cells	0–2/high-power field	
White blood cells	0–2/high-power field	
Bacteria	None	
Crystals	None	
Bladder cells	None	
Squamous cells	None	
Tubular cells	None	
Broad casts	None	
Epithelial cell casts	None	
Granular casts	None	
Hyaline casts	0–5/low-power field	
Red blood cell casts	None	
Waxy casts	None	
White cell casts	None	
Sodium (varies with intake)	100–260 mmol/d	100–260 meq/d
Specific gravity:		
After 12-h fluid restriction	>1.025	>1.025
After 12-h deliberate water intake	≤1.003	≤1.003
Tubular reabsorption, phosphorus	0.79–0.94 of filtered load	79–94% of filtered load
Urea nitrogen	214–607 mmol/d	6–17 g/d
Uric acid (normal diet)	1.49–4.76 mmol/d	250–800 mg/d
Vanillylmandelic acid (VMA)	<30 µmol/d	<6 mg/d

TABLE 10

NORMAL PRESSURES IN HEART AND GREAT VESSELS

PRESSURE (mmHg)	AVERAGE	RANGE
Right Atrium		
Mean	2.8	1–5
a wave	5.6	2.5–7
c wave	3.8	1.5–6
x wave	1.7	0–5
v wave	4.6	2–7.5
y wave	2.4	0–6
Right Ventricle		
Peak systolic	25	17–32
End-diastolic	4	1–7
Pulmonary Artery		
Mean	15	9–19
Peak systolic	25	17–32
End-diastolic	9	4–13
Pulmonary Artery Wedge		
Mean	9	4.5–13
Left Atrium		
Mean	7.9	2–12
a wave	10.4	4–16
v wave	12.8	6–21
Left Ventricle		
Peak systolic	130	90–140
End-diastolic	8.7	5–12
Brachial Artery		
Mean	85	70–105
Peak systolic	130	90–140
End-diastolic	70	60–90

Source: Reproduced from: MJ Kern *The Cardiac Catheterization Handbook*, 4th ed. Philadelphia, Mosby, 2003.

TABLE 11

CIRCULATORY FUNCTION TESTS

TEST	RESULTS: REFERENCE RANGE	
	SI UNITS (RANGE)	CONVENTIONAL UNITS (RANGE)
Arteriovenous oxygen difference	30–50 mL/L	30–50 mL/L
Cardiac output (Fick)	2.5–3.6 L/m ² of body surface area per min	2.5–3.6 L/m ² of body surface area per min
Contractility indexes		
Max. left ventricular dp/dt (dp/dt)	220 kPa/s (176–250 kPa/s)	1650 mmHg/s (1320–1880 mmHg/s)
DP when DP = 5.3 kPa	(37.6 ± 12.2)/s	(37.6 ± 12.2)/s
(40 mmHg) (DP, developed LV pressure)	3.32 ± 0.84 end-diastolic volumes per second	3.32 ± 0.84 end-diastolic volumes per second
Mean normalized systolic ejection rate (angiography)	1.83 ± 0.56 circumferences per second	1.83 ± 0.56 circumferences per second
Mean velocity of circumferential fiber shortening (angiography)		
Ejection fraction: stroke volume/end-diastolic volume (SV/EDV)	0.67 ± 0.08 (0.55–0.78)	0.67 ± 0.08 (0.55–0.78)
End-diastolic volume	70 ± 20.0 mL/m ² (60–88 mL/m ²)	70 ± 20.0 mL/m ² (60–88 mL/m ²)
End-systolic volume	25 ± 5.0 mL/m ² (20–33 mL/m ²)	25 ± 5.0 mL/m ² (20–33 mL/m ²)
Left ventricular work		
Stroke work index	50 ± 20.0 (g·m)/m ² (30–110)	50 ± 20.0 (g·m)/m ² (30–110)
Left ventricular minute work index	1.8–6.6 [(kg·m)/m ²]/min	1.8–6.6 [(kg·m)/m ²]/min
Oxygen consumption index	110–150 mL	110–150 mL
Maximum oxygen uptake	35 mL/min (20–60 mL/min)	35 mL/min (20–60 mL/min)
Pulmonary vascular resistance	2–12 (kPa·s)/L	20–130 (dyn·s)/cm ⁵
Systemic vascular resistance	77–150 (kPa·s)/L	770–1600 (dyn·s)/cm ⁵

Source: E Braunwald et al: *Heart Disease*, 6th ed. Philadelphia, W.B. Saunders Co., 2001.

TABLE 12

NORMAL ECHOCARDIOGRAPHIC REFERENCE LIMITS AND PARTITION VALUES IN ADULTS

	WOMEN REFERENCE RANGE	MILDLY ABNORMAL	MODERATELY ABNORMAL	SEVERELY ABNORMAL	MEN REFERENCE RANGE	MILDLY ABNORMAL	MODERATELY ABNORMAL	SEVERELY ABNORMAL
Left ventricular dimensions								
Septal thickness, cm	0.6–0.9	1.0–1.2	1.3–1.5	≥1.6	0.6–1.0	1.1–1.3	1.4–1.6	≥1.7
Posterior wall thickness, cm	0.6–0.9	1.0–1.2	1.3–1.5	≥1.6	0.6–1.0	1.1–1.3	1.4–1.6	≥1.7
Diastolic diameter, cm	3.9–5.3	5.4–5.7	5.8–6.1	≥6.2	4.2–5.9	6.0–6.3	6.4–6.8	≥6.9
Diastolic diameter/BSA, cm/m ²	2.4–3.2	3.3–3.4	3.5–3.7	≥3.8	2.2–3.1	3.2–3.4	3.5–3.6	≥3.7
Diastolic diameter/height, cm/m	2.5–3.2	3.3–3.4	3.5–3.6	≥3.7	2.4–3.3	3.4–3.5	3.6–3.7	≥3.8
Left ventricular volumes								
Diastolic, mL	56–104	105–117	118–130	≥131	67–155	156–178	179–201	≥202
Diastolic/BSA, mL/m ²	35–75	76–86	87–96	≥97	35–75	76–86	87–96	≥97
Systolic, mL	19–49	50–59	60–69	≥70	22–58	59–70	71–82	≥83
Systolic/BSA, mL/m ²	12–30	31–36	37–42	≥43	12–30	31–36	37–42	≥43
Left ventricular mass, 2D method								
Mass, g	66–150	151–171	172–182	≥183	96–200	201–227	228–254	≥255
Mass/BSA, g/m ²	44–88	89–100	101–112	≥113	50–102	103–116	117–130	≥131
Left ventricular function								
Endocardial fractional shortening (%)	27–45	22–26	17–21	≤16	25–43	20–24	15–19	≤14
Midwall fractional shortening (%)	15–23	13–14	11–12	≤10	14–22	12–13	10–11	≤9
Ejection fraction, 2D method (%)	≥55	45–54	30–44	≤29	≥55	45–54	30–44	≤29
Right heart dimensions (cm)								
Basal RV diameter	2.0–2.8	2.9–3.3	3.4–3.8	≥3.9	2.0–2.8	2.9–3.3	3.4–3.8	≥3.9
Mid-RV diameter	2.7–3.3	3.4–3.7	3.8–4.1	≥4.2	2.7–3.3	3.4–3.7	3.8–4.1	≥4.2
Base-to-apex length	7.1–7.9	8.0–8.5	8.6–9.1	≥9.2	7.1–7.9	8.0–8.5	8.6–9.1	≥9.2
RVOT diameter above aortic valve	2.5–2.9	3.0–3.2	3.3–3.5	≥3.6	2.5–2.9	3.0–3.2	3.3–3.5	≥3.6
RVOT diameter above pulmonic valve	1.7–2.3	2.4–2.7	2.8–3.1	≥3.2	1.7–2.3	2.4–2.7	2.8–3.1	≥3.2
Pulmonary artery diameter below pulmonic valve	1.5–2.1	2.2–2.5	2.6–2.9	≥3.0	1.5–2.1	2.2–2.5	2.6–2.9	≥3.0
Right ventricular size and function in 4-chamber view								
Diastolic area, cm ²	11–28	29–32	33–37	≥38	11–28	29–32	33–37	≥38
Systolic area, cm ²	7.5–16	17–19	20–22	≥23	7.5–16	17–19	20–22	≥23
Fractional area change, %	32–60	25–31	18–24	≤17	32–60	25–31	18–24	≤17
Atrial sizes								
LA diameter, cm	2.7–3.8	3.9–4.2	4.3–4.6	≥4.7	3.0–4.0	4.1–4.6	4.7–5.2	≥5.3
LA diameter/BSA, cm/m ²	1.5–2.3	2.4–2.6	2.7–2.9	≥3.0	1.5–2.3	2.4–2.6	2.7–2.9	≥3.0
RA minor axis, cm	2.9–4.5	4.6–4.9	5.0–5.4	≥5.5	2.9–4.5	4.6–4.9	5.0–5.4	≥5.5
RA minor axis/BSA, cm/m ²	1.7–2.5	2.6–2.8	2.9–3.1	≥3.2	1.7–2.5	2.6–2.8	2.9–3.1	≥3.2

(continued)

TABLE 12

NORMAL ECHOCARDIOGRAPHIC REFERENCE LIMITS AND PARTITION VALUES IN ADULTS (CONTINUED)

	WOMEN REFERENCE RANGE	MILDLY ABNORMAL	MODERATELY ABNORMAL	SEVERELY ABNORMAL	MEN REFERENCE RANGE	MILDLY ABNORMAL	MODERATELY ABNORMAL	SEVERELY ABNORMAL
LA area, cm ²	<20	20–30	30–40	≥41	<20	20–30	30–40	≥41
LA volume, mL	22–52	53–62	63–72	≥73	18–58	59–68	69–78	≥79
LA volume/BSA, mL/m ²	16–28	29–33	34–39	≥40	16–28	29–33	34–39	≥40
Aortic stenosis, classification of severity								
Aortic jet velocity, m/s		2.6–2.9	3.0–4.0	>4.0		2.6–2.9	3.0–4.0	>4.0
Mean gradient, mmHg		<20	20–40	>40		<20	20–40	>40
Valve area, cm ²		>1.5	1.0–1.5	<1.0		>1.5	1.0–1.5	<1.0
Indexed valve area, cm ² /m ²		>0.85	0.60–0.85	<0.6		>0.85	0.60–0.85	<0.6
Velocity ratio		>0.50	0.25–0.50	<0.25		>0.50	0.25–0.50	<0.25
Mitral stenosis, classification of severity								
Valve area, cm ²		>1.5	1.0–1.5	<1.0		>1.5	1.0–1.5	<1.0
Mean gradient, mmHg		<5	5–10	>10		<5	5–10	>10
Pulmonary artery pressure, mmHg		<30	30–50	>50		<30	30–50	>50
Aortic regurgitation, indices of severity								
Vena contracta width, cm		<0.30	0.30–0.60	≥0.60		<0.30	0.30–0.60	≥0.60
Jet width/LVOT width, %		<25	25–64	≥65		<25	25–64	≥65
Jet CSA/LVOT CSA, %		<5	5–59	≥60		<5	5–59	≥60
Regurgitant volume, mL/beat		<30	30–59	≥60		<30	30–59	≥60
Regurgitant fraction, %		<30	30–49	≥50		<30	30–49	≥50
Effective regurgitant orifice area, cm ²		<0.10	0.10–0.29	≥0.30		<0.10	0.10–0.29	≥0.30
Mitral regurgitation, indices of severity								
Vena contracta width, cm		<0.30	0.30–0.69	≥0.70		<0.30	0.30–0.69	≥0.70
Regurgitant volume, mL/beat		<30	30–59	≥60		<30	30–59	≥60
Regurgitant fraction, %		<30	30–49	≥50		<30	30–49	≥50
Effective regurgitant orifice area, cm ²		<0.20	0.20–0.39	≥0.40		<0.20	0.20–0.39	≥0.40

Abbreviations: BSA, body surface area; CSA, cross-sectional area; LA, left atrium; LVOT, left ventricular outflow tract; RA, right atrium; RV, right ventricle; RVOT, right ventricular outflow tract; 2D, 2-dimensional.

Source: Values adapted from: American Society of Echocardiography, Guidelines and Standards. <http://www.asecho.org/i4a/pages/index.cfm?pageid=3317>. Accessed Feb 23, 2010.

TABLE 13

SUMMARY OF VALUES USEFUL IN PULMONARY PHYSIOLOGY

		TYPICAL VALUES	
	SYMBOL	MAN AGED 40, 75 kg, 175 cm TALL	WOMAN AGED 40, 60 kg, 160 cm TALL
Pulmonary Mechanics			
Spirometry—volume-time curves			
Forced vital capacity	FVC	5.0 L	3.4 L
Forced expiratory volume in 1 s	FEV ₁	4.0 L	2.8 L
FEV ₁ /FVC	FEV ₁ %	80%	78%
Maximal midexpiratory flow rate	MMEF (FEF 25–75)	4.1 L/s	3.2 L/s
Maximal expiratory flow rate	MEFR (FEF 200–1200)	9.0 L/s	6.1 L/s
Spirometry—flow-volume curves			
Maximal expiratory flow at 50% of expired vital capacity	V _{max} 50 (FEF 50%)	5.0 L/s	4.0 L/s
Maximal expiratory flow at 75% of expired vital capacity	V _{max} 75 (FEF 75%)	2.1 L/s	2.0 L/s
Resistance to airflow:			
Pulmonary resistance	RL (R _L)	<3.0 (cmH ₂ O/s)/L	
Airway resistance	R _{aw}	<2.5 (cmH ₂ O/s)/L	
Specific conductance	SG _{aw}	>0.13 cmH ₂ O/s	
Pulmonary compliance			
Static recoil pressure at total lung capacity	P _{st} TLC	25 ± 5 cmH ₂ O	
Compliance of lungs (static)	CL	0.2 L cmH ₂ O	
Compliance of lungs and thorax	C(L + T)	0.1 L cmH ₂ O	
Dynamic compliance of 20 breaths per minute	C _{dyn} 20	0.25 ± 0.05 L/cmH ₂ O	
Maximal static respiratory pressures:			
Maximal inspiratory pressure	MIP	>110 cmH ₂ O	>70 cmH ₂ O
Maximal expiratory pressure	MEP	>200 cmH ₂ O	>140 cmH ₂ O
Lung Volumes			
Total lung capacity	TLC	6.9 L	4.9 L
Functional residual capacity	FRC	3.3 L	2.6 L
Residual volume	RV	1.9 L	1.5 L
Inspiratory capacity	IC	3.7 L	2.3 L
Expiratory reserve volume	ERV	1.4 L	1.1 L
Vital capacity	VC	5.0 L	3.4 L
Gas Exchange (Sea Level)			
Arterial O ₂ tension	P _{aO₂}	12.7 ± 0.7 kPa (95 ± 5 mmHg)	
Arterial CO ₂ tension	P _{aCO₂}	5.3 ± 0.3 kPa (40 ± 2 mmHg)	
Arterial O ₂ saturation	S _{aO₂}	0.97 ± 0.02 (97 ± 2%)	
Arterial blood pH	pH	7.40 ± 0.02	
Arterial bicarbonate	HCO ₃ ⁻	24 + 2 meq/L	
Base excess	BE	0 ± 2 meq/L	
Diffusing capacity for carbon monoxide (single breath)	DL _{CO}	37 mL CO/min/mmHg	27 mL CO/min/mmHg
Dead space volume	V _D	2 mL/kg body wt	
Physiologic dead space; dead space-tidal volume ratio	V _D /V _T		
Rest		≤35% V _T	
Exercise		≤20% V _T	
Alveolar-arterial difference for O ₂	P(A – a) _{O₂}	≤2.7 kPa ≤20 kPa (≤24 mmHg)	

Source: Based on: AH Morris et al: *Clinical Pulmonary Function Testing. A Manual of Uniform Laboratory Procedures*, 2nd ed. Salt Lake City, Utah, Intermountain Thoracic Society, 1984.

TABLE 14
GASTROINTESTINAL TESTS

TEST	RESULTS	
	SI UNITS	CONVENTIONAL UNITS
Absorption tests		
D-Xylose: after overnight fast, 25 g xylose given in oral aqueous solution		
Urine, collected for following 5 h	25% of ingested dose	25% of ingested dose
Serum, 2 h after dose	2.0–3.5 mmol/L	30–52 mg/dL
Vitamin A: a fasting blood specimen is obtained and 200,000 units of vitamin A in oil is given orally	Serum level should rise to twice fasting level in 3–5 h	Serum level should rise to twice fasting level in 3–5 h
Bentiromide test (pancreatic function): 500 mg bentiromide (chymex) orally; <i>p</i> -aminobenzoic acid (PABA) measured		
Plasma		>3.6 (±1.1) µg/mL at 90 min
Urine	>50% recovered in 6 h	>50% recovered in 6 h
Gastric juice		
Volume		
24 h	2–3 L	2–3 L
Nocturnal	600–700 mL	600–700 mL
Basal, fasting	30–70 mL/h	30–70 mL/h
Reaction		
pH	1.6–1.8	1.6–1.8
Titratable acidity of fasting juice	4–9 µmol/s	15–35 meq/h
Acid output		
Basal		
Females (mean ± 1 SD)	0.6 ± 0.5 µmol/s	2.0 ± 1.8 meq/h
Males (mean ± 1 SD)	0.8 ± 0.6 µmol/s	3.0 ± 2.0 meq/h
Maximal (after SC histamine acid phosphate, 0.004 mg/kg body weight, and preceded by 50 mg promethazine, or after betazole, 1.7 mg/kg body weight, or pentagastrin, 6 µg/kg body weight)		
Females (mean ± 1 SD)	4.4 ± 1.4 µmol/s	16 ± 5 meq/h
Males (mean ± 1 SD)	6.4 ± 1.4 µmol/s	23 ± 5 meq/h
Basal acid output/maximal acid output ratio	≤0.6	≤0.6
Gastrin, serum	0–200 µg/L	0–200 pg/mL
Secretin test (pancreatic exocrine function): 1 unit/kg body weight, IV		
Volume (pancreatic juice) in 80 min	>2.0 mL/kg	>2.0 mL/kg
Bicarbonate concentration	>80 mmol/L	>80 meq/L
Bicarbonate output in 30 min	>10 mmol	>10 meq

TABLE 15

BODY FLUIDS AND OTHER MASS DATA

	REFERENCE RANGE	
	SI UNITS	CONVENTIONAL UNITS
Ascitic fluid		
Body fluid		
Total volume (lean) of body weight	50% (in obese) to 70%	
Intracellular	30–40% of body weight	
Extracellular	20–30% of body weight	
Blood		
Total volume		
Males	69 mL/kg body weight	
Females	65 mL/kg body weight	
Plasma volume		
Males	39 mL/kg body weight	
Females	40 mL/kg body weight	
Red blood cell volume		
Males	30 mL/kg body weight	1.15–1.21 L/m ² of body surface area
Females	25 mL/kg body weight	0.95–1.00 L/m ² of body surface area
Body mass index	18.5–24.9 kg/m ²	18.5–24.9 kg/m ²

TABLE 16

RADIATION-DERIVED UNITS

QUANTITY	MEASURES	OLD UNIT	SI UNIT	SPECIAL NAME FOR SI UNIT (ABBREVIATION)	CONVERSION
Activity	Rate of radioactive decay	curie (Ci)	Disintegrations per second (dps)	becquerel (Bq)	1 Ci = 3.7×10^{10} Bq 1 mCi = 37 MBq 1 Bq = 2.703×10^{-11} Ci
Exposure	Amount of ionizations produced in dry air by x-rays or gamma rays, per unit of mass	roentgen (R)	Coulomb per kilogram (C/kg)	none	1 C/kg = 3876 R 1 R = 2.58×10^{-4} C/kg 1 mR = 258 pC/kg
Air kerma	Sum of initial energies of charged particles liberated by ionizing radiation in air, per unit of mass	rad	Joule per kilogram (J/kg)	gray (Gy)	1 Gy = 100 rad 1 rad = 0.01 Gy 1 mrad = 10 μ Gy
Absorbed dose	Energy deposited per unit of mass in a medium, e.g., an organ/tissue	rad	Joule per kilogram (J/kg)	gray (Gy)	1 Gy = 100 rad 1 rad = 0.01 Gy 1 mrad = 10 μ Gy
Equivalent dose	Energy deposited per unit of mass in a medium, e.g., an organ/tissue, weighted to reflect type(s) of radiation	rem	Joule per kilogram (J/kg)	sievert (Sv)	1 Sv = 100 rem 1 rem = 0.01 Sv 1 mrem = 10 μ Sv
Effective dose	Energy deposited per unit of mass in a reference individual, doubly weighted to reflect type(s) of radiation and organ(s) irradiated	rem	Joule per kilogram (J/kg)	sievert (Sv)	1 Sv = 100 rem 1 rem = 0.01 Sv 1 mrem = 10 μ Sv

and T_4 level is unknown presently, so thyroid uptake scan is not indicated at this time.

74. The answer is A.

(Chap. 38) The diagnosis of Cushing's syndrome relies on documentation of endogenous hypercortisolism. Of the list of choices, the most cost-effective and precise test is the 24-hour urine free cortisol. Failure to suppress plasma morning cortisol after overnight suppression with 1 mg dexamethasone is an alternative. Most ACTH-secreting pituitary adenomas are less than 5 mm in diameter, and approximately half are not detected even with sensitive MRI. Further, because incidental microadenomas are common in the pituitary, the presence of a small pituitary abnormality on MRI may not establish the source of ACTH production. Basal plasma ACTH levels are used to distinguish between ACTH-independent (adrenal or exogenous glucocorticoid) and ACTH-dependent (pituitary, ectopic ACTH) sources of hypercortisolism. Mean basal ACTH levels are higher in patients with ectopic ACTH production than in patients with pituitary ACTH adenomas. There is significant overlap in ACTH levels, however, and this test should not be used as an initial diagnostic test. Rarely, patients have Cushing's syndrome and elevated ACTH due to a CRH-releasing tumor. In this case, CRH levels are elevated. Inferior petrosal venous sampling can be used to identify a pituitary source of ACTH secretion when imaging modalities do not reveal a source.

75. The answer is B.

(Chap. 38) The identification of an empty sella is often the result of an incidental MRI finding. Typically these patients will have normal pituitary function and should be reassured. It is likely that the surrounding rim of pituitary tissue is functioning normally. An empty sella may signal the insidious onset of hypopituitarism, and laboratory results should be followed closely. Unless her clinical situation changes, repeat MRI is not indicated. Endocrine malignancy is unlikely, and surgery is not part of the management of an empty sella.

76. The answer is D.

(Chap. 39) The onset of multiple sclerosis (MS) may be abrupt or insidious. Symptoms may be severe or seem so trivial that a patient may not seek medical attention for months or years. Indeed, at autopsy, approximately 0.1% of individuals who were asymptomatic during life will be found, unexpectedly, to have pathologic evidence of MS. Similarly, in the modern era, an MRI scan obtained for an unrelated reason may show evidence of asymptomatic MS. Symptoms of MS are extremely varied and depend on the location and severity of lesions within the CNS (see Table 39-2). Examination often reveals evidence of neurologic dysfunction, often in asymptomatic locations. For example, a patient may present with symptoms in one leg but signs in both.

77. The answer is D.

(Chap. 39) The four clinical types of multiple sclerosis (MS) include relapsing/remitting, secondary progressive, primary progressive, and progressive relapsing. Relapsing/remitting MS (RRMS) accounts for 85% of MS cases at onset and is characterized by discrete attacks that generally evolve over days to weeks (rarely over hours). There is often complete recovery over the ensuing weeks to months. However, when ambulation is severely impaired during an attack, approximately half will fail to improve. Between attacks, patients are neurologically stable. Secondary progressive MS (SPMS) always begins as RRMS. At some point, however, the clinical course changes so that the patient experiences a steady deterioration in function unassociated with acute attacks (which may continue or cease during the progressive phase). SPMS produces a greater amount of fixed neurologic disability than RRMS. For a patient with RRMS, the risk of developing SPMS is approximately 2% each year, meaning that the great majority of RRMS ultimately evolves into SPMS. SPMS appears to represent a late stage of the same underlying illness as RRMS. Primary progressive MS (PPMS) accounts for approximately 15% of cases. These patients do not experience attacks but only a steady functional decline from disease onset. Compared to RRMS, the sex distribution is more even, the disease begins later in life (mean age approximately 40 years), and disability develops faster (at least relative to the onset of the first clinical symptom). Despite these differences, PPMS appears to represent the same underlying illness as RRMS. Progressive/relapsing MS (PRMS) overlaps PPMS and SPMS and accounts for about 5% of MS patients. Like patients with PPMS, these patients experience a steady deterioration in their condition from disease onset. However, like SPMS patients, they experience occasional attacks superimposed upon their progressive course. Autoimmune autonomic neuropathy is a distinct clinical syndrome not related to MS. It presents with the subacute development of autonomic disturbances with orthostatic hypotension, enteric neuropathy (gastroparesis, ileus, constipation/diarrhea), and cholinergic failure; the latter consists of loss of sweating, sicca complex, and a tonic pupil. Autoantibodies against the ganglionic ACh receptor (A_3 AChR) are present in the serum of many patients and are now considered to be diagnostic of this syndrome.

78. The answer is D.

(Chap. 40) In a patient with suspected bacterial meningitis empirical therapy should be administered promptly to reduce mortality and morbidity. The decision to obtain an imaging study prior to lumbar puncture (LP) is based on the concern of precipitating herniation in a patient with elevated intracranial pressure or focal CNS lesions. Therefore, patients with the presence of papilledema on physical examination, history of recent head trauma, known or suspected intracranial lesions (immunosuppressed, known

malignancy), focal neurologic findings, or depressed level of consciousness should have a head CT or MRI prior to LP. In an immunocompetent patient with no known history of recent head trauma, a normal level of consciousness, and no evidence of papilledema or focal neurologic deficits, it is considered safe to perform LP without prior neuroimaging studies. Kernig's sign is elicited in a supine patient by flexing the thigh and knee. A positive sign occurs when the patient has head/neck pain when passively straightening the knee. The sensitivity and specificity of this sign (also Brudzinski's) for bacterial meningitis are unknown, but they imply meningeal irritation, not an intracranial lesion or elevated intracranial pressure. While cerebrospinal fluid cultures may be impacted by administration of antibiotics prior to LP, stains, antigen tests, and polymerase chain reaction tests will not be affected.

79. The answer is B.

(Chap. 40) The release of bacterial cell wall components after killing by antibiotics may evoke a marked inflammatory cytokine response in the subarachnoid space. This inflammation may lead to increased damage of the blood-brain barrier and central nervous system damage. Glucocorticoids can blunt this response by inhibiting tumor necrosis factor and interleukin-1. They work best if administered before antibiotics. Clinical trials have demonstrated that dexamethasone, 10 mg IV administered 20 minutes *before* antibiotics, reduced unfavorable outcomes, including death. The dexamethasone was continued for 4 days. The benefits were most striking in pneumococcal meningitis. Because this is the most common cause of meningitis in the elderly, empirical coverage should include this intervention as well. The efficacy of dexamethasone therapy in preventing neurologic sequelae is different between high- and low-income countries. Randomized trials in low-income countries (sub-Saharan Africa, Southeast Asia) failed to show benefit in subgroups of patients. The lack of efficacy of dexamethasone in these trials has been attributed to late presentation to the hospital with more advanced disease, antibiotic pretreatment, malnutrition, infection with HIV, and treatment of patients with probable, but not microbiologically proven, bacterial meningitis. The results of these clinical trials suggest that patients in sub-Saharan Africa and those in low-income countries with negative CSF Gram stain and culture should not be treated with dexamethasone. Empirical antibiotics in this case should include a third-generation cephalosporin, vancomycin, and ampicillin. However, dexamethasone may decrease vancomycin penetration into the CSF, so its use should be considered carefully in cases where the most likely organism requires vancomycin coverage. Acyclovir or valacyclovir may be used as initial empiric treatment in cases of suspected herpes CNS infection. However, in this case the LP is highly suggestive of acute bacterial infection. Intravenous gamma globulin is used as adjunctive therapy in children with known

immunoglobulin deficiency who are at risk of viral meningitis/encephalitis.

80. The answer is D.

(Chap. 40) *Listeria* has become an increasingly important cause of bacterial meningitis in neonates (<1 month of age), pregnant women, individuals more than 60 years old, and immunocompromised individuals. Infection is acquired by eating contaminated foods such as unpasteurized dairy products, coleslaw, milk, soft cheeses, delicatessen meats, and uncooked hot dogs. Ampicillin is the agent most often added to the initial empirical regimen to cover *L. monocytogenes*.

81. The answer is D.

(Chap. 41) Ibuprofen, isoniazid, ciprofloxacin, tolmetin, sulfa-containing medicines, and phenazopyridine have been implicated in drug hypersensitivity leading to meningitis. The cerebrospinal fluid (CSF) will typically show neutrophils, but mononuclear cells or eosinophils are occasionally present. Most causes of chronic (not recurrent) meningitis cause a predominance of mononuclear cells. The differential for chronic meningitis is broad and a diagnosis is often difficult to make. The treating physician needs to consider a diverse array of viral, fungal, bacterial, mycobacterial, helminthic, and protozoal pathogens, both common and exotic, and therefore should obtain a detailed social history and consult an expert in the field. Recurrent meningitis is often due to herpes simplex virus type 2 infection and this should be ruled out, particularly if active genital ulcers develop concurrently. Malignancy, sarcoidosis, and vasculitis are all potential causes, and history, physical examination, and appropriate further testing should dictate the degree to which these possibilities are explored. Medications are often overlooked as a cause of chronic meningitis and should always be carefully considered. When CSF neutrophils predominate after 3 weeks of illness, *Nocardia*, *Actinomyces*, *Brucella*, tuberculosis (<10% of cases), and fungal and noninfectious causes of chronic meningitis should be considered.

82. The answer is E.

(Chap. 43) Prions are infectious particles that cause central nervous system degeneration. The human prion diseases described to date include Creutzfeldt-Jacob disease (CJD), kuru, Gerstmann-Sträussler-Scheinker disease, and fatal insomnia. The most common prion disease is sporadic CJD (sCJD), which occurs in a seemingly random pattern in adults in their fifth and sixth decades of life. sCJD accounts for about 85% of cases of CJD and occurs in approximately 1 per 1 million population. Variant CJD (vCJD) results from infection from bovine exposure to tainted beef from cattle with bovine spongiform encephalopathy (BSE). There has been a steady decline of cases of vCJD in Europe over the past decade. Infectious CJD (iCJD) has resulted from injection of tainted human growth hormone, as well

as transplant of infected dura mater grafts into humans. Familial CJD (fCJD) is due to germ-line mutations that follow an autosomal dominant inheritance. Kuru is due to infection through ritualistic cannibalism. Gerstmann-Sträussler-Scheinker disease and familial fatal insomnia (FFI) occur as dominantly inherited prion diseases. Sporadic cases of fatal insomnia (sFI) have been described.

83. The answer is B.

(Chap. 43) Startle myoclonus is a worrisome sign but is neither sensitive nor specific for CJD, though it is more worrisome if it occurs during sleep. The constellation of dementia, myoclonus, and periodic electrical bursts in an afebrile 60-year-old patient generally indicates CJD. Clinical abnormalities in CJD are confined to the CNS. Lewy body dementia, Alzheimer's disease, central nervous system infections, and myoclonic epilepsy can all cause myoclonus. Both EEG and MRI can help differentiate CJD from these disorders. The MRI finding of cortical ribboning and intensity in the basal ganglia on fluid-attenuated inversion recovery sequences is characteristic of CJD. EEG is useful if stereotypical periodic bursts every 1–2 seconds are present, but this is seen in only 60% of cases, and other findings may be less specific. Demonstration of specific immunoassays for proteolytic products of disease-causing prion proteins (PrP^{Sc}) at brain biopsy may be necessary to confirm diagnosis in some cases. However, these proteins are not uniformly distributed throughout the brain and false-negative biopsies occur. Both surgeons and pathologists must be warned to use standard precautions under these circumstances. These proteins cannot be measured from cerebrospinal fluid (CSF). CSF in CJD is usually normal except for a minimally elevated protein. Many patients with CJD have elevated CSF stress protein 14-3-3. This test alone is neither sensitive nor specific, as patients with herpes simplex virus encephalitis, multi-infarct dementia, and stroke may have similar elevations.

84. The answer is C.

(Chap. 44) One of the better characterized paraneoplastic neurologic syndromes is cerebellar ataxia caused by Purkinje cell drop-out in the cerebellum; it is manifested by dysarthria, limb and gait ataxia, and nystagmus. Radiologic imaging reveals cerebellar atrophy. Many antibodies have been associated with this syndrome, including anti-Yo, anti-Tr, and antibodies to the glutamate receptor. Although lung cancer, particularly small cell cancer, accounts for a large number of patients with neoplasm-associated cerebellar ataxia, those with the syndrome who display anti-Yo antibodies in the serum typically have breast or ovarian cancer. Cerebellar ataxia may also be seen in Hodgkin's lymphoma in association with anti-Tr antibodies.

85. The answer is A.

(Chap. 45) Charcot-Marie-Tooth (CMT) syndrome is the most common type of hereditary neuropathy. CMT

is comprised of several similar but genetically distinct conditions with different associated mutations. CMT1 is the most common and is an inherited demyelinating sensorimotor neuropathy. CMT1 affects patients in the first to third decades of life with distal leg weakness, i.e., footdrop. Although patients generally do not complain of sensory symptoms, these can often be elicited on physical examination. Muscle stretch reflexes are unobtainable or reduced throughout and calves are often atrophied, which makes legs appear to have a so-called inverted champagne bottle appearance. Hereditary neuralgic amyotrophy (HNA) is an autosomal dominant disorder characterized by recurrent attacks of pain, weakness, and sensory loss in the distribution of the brachial plexus that often begins in childhood. Hereditary sensory and autonomic neuropathy (HSAN) is a very rare group of hereditary neuropathies in which sensory and autonomic dysfunction predominates over muscle weakness. This would not fit the clinical pattern described here. Guillain-Barré syndrome presents generally acutely with involvement of both proximal and distal weakness and sensory loss. The prolonged symptom period and distribution described here is not typical for Guillain-Barré syndrome. Fabry disease is an X-linked disorder in which men are more commonly affected than women. Patients have angiokeratomas, which are reddish-purple lesions usually found around the umbilicus, scrotum, and inguinal region. Burning pain in the hands and feet often is found in late childhood or early adult life. Patients also have premature atherosclerosis from the underlying mutation in the alpha-galactosidase gene with accumulation of ceramide in nerves and blood vessels.

86. The answer is E.

(Chap. 45) One of the most common side effects of isoniazid treatment is peripheral neuropathy. The elderly, malnourished, and "slow acetylators" are at increased risk for developing the neuropathy. INH inhibits pyridoxal phosphokinase, resulting in pyridoxine (vitamin B₆) deficiency and the neuropathy. Prophylactic administration of pyridoxine can prevent the neuropathy from developing. Symptoms are generally dysesthesias and sensory ataxia. Impaired large-fiber sensory modalities are found on examination. Cobalamin (B₁₂) is not reduced in this condition and is unaffected by isoniazid. Neurontin and pregabalin may alleviate symptoms but will not reverse the neuropathy. There is no indication that hypothyroidism is present.

87. The answer is C.

(Chap. 45) Diabetes mellitus (DM) is the most common cause of peripheral neuropathy in developed countries and is associated with several different types of polyneuropathy including distal symmetric sensory or sensorimotor polyneuropathy, autonomic neuropathy, diabetic neuropathic cachexia, polyradiculoneuropathies, cranial neuropathies, and other mononeuropathies. Risk factors for the

development of neuropathy include long-standing and poorly controlled diabetes and the presence of retinopathy or nephropathy. The patient here appears to have diabetic distal symmetric sensory and sensorimotor polyneuropathy (DSPN), which is the most common form of diabetic neuropathy and presents with sensory loss beginning in the toes and gradually progresses over time up the legs and into the fingers and arms. Symptoms also may include tingling, burning, and deep, aching pains. Nerve biopsy, though rarely indicated, often shows axonal degeneration, endothelial hyperplasia, and occasionally perivascular inflammation. Tight glucose control prevents the development of disease but does not reverse established disease. Diabetic autonomic neuropathy is often seen in combination with DSPN and manifests by abnormal sweating, dysfunctional thermoregulation, dry eyes and mouth, postural hypotension, gastrointestinal abnormalities including gastroparesis, and genitourinary dysfunction.

88. The answer is B.

(Chap. 45) Peripheral neuropathy is a general term indicating peripheral nerve disorders of any cause. The causes are legion, but peripheral neuropathy can be classified by a number of means: axonal versus demyelinating, mononeuropathy versus polyneuropathy versus mononeuritis multiplex, sensory versus motor, and by the tempo of the onset of symptoms. Mononeuropathy typically results from local compression, trauma, or entrapment of a nerve. Polyneuropathy often results from a more systemic process. The distinction between axonal and demyelinating can often be made only with nerve conduction studies. HIV infection causes a common, distal, symmetric, mainly sensory polyneuropathy. Vitamin B₁₂ deficiency typically causes a sensory neuropathy that predominantly involves the dorsal columns. Hypothyroidism and acromegaly may both cause compression and swelling of nerve fibers, resulting first in sensory symptoms and later in disease with motor symptoms. Critical illness polyneuropathy is predominantly motor in presentation. Patients typically present with weakness that can be profound. These patients may recover over the course of weeks to months. The etiology is unknown, but an association may exist with prolonged immobilization, severity of illness, neuromuscular blockade, and corticosteroids.

89. The answer is B.

(Chap. 45) Carpal tunnel syndrome is caused by the entrapment of the median nerve at the wrist. Symptoms begin with paresthesias in the median nerve distribution. With worsening, atrophy and weakness may develop. This condition is most commonly caused by excessive use of the wrist and situations involving repetitive motion. Most cases are idiopathic other than those related to occupational or environmental associations. Less commonly, systemic disease may result in carpal tunnel syndrome related to nerve compression or infiltrative disease.

This may be suspected when bilateral disease is apparent. Tenosynovitis with arthritis, as in the case of rheumatoid arthritis, and thickening of the connective tissue, as in the case of amyloid or acromegaly, may cause carpal tunnel syndrome. Other systemic diseases, such as hypothyroidism and diabetes mellitus, are also possible etiologies. Acute or chronic leukemia is not typically associated with carpal tunnel syndrome.

90. The answer is B.

(Chap. 46) Guillain-Barré syndrome (GBS) is an acute, severe polyradiculoneuropathy that is autoimmune in nature. GBS manifests as rapidly evolving, areflexic motor paralysis with or without sensory disturbance, usually with ascending paralysis developing over several days. Approximately 70% of GBS cases occur 1–3 weeks after an acute infectious process, usually respiratory or gastrointestinal. Twenty to thirty percent of cases in North America, Europe, and Australia are preceded by infection or reinfection with *Campylobacter jejuni*. Other implicated infections include Epstein-Barr virus, CMV, and *Mycoplasma pneumoniae*. *T. whippelii* is the etiologic agent of Whipple's disease and *B. henselae* is implicated in cat-scratch fever.

91. The answer is E.

(Chap. 47) Myasthenia gravis (MG) is a neuromuscular disorder characterized by weakness and fatigability of skeletal muscles. The primary defect is a decrease in the number of acetylcholine receptors at the neuromuscular junction secondary to autoimmune antibodies. MG is not rare, affecting at least 1 in 7500 individuals. Women are affected more frequently than men. Women typically present in the second and third decades of life, and men present in the fifth and sixth decades. The key features of MG are weakness and fatigability. Clinical features include weakness of the cranial muscles, particularly the eyelids and extraocular muscles. Diplopia and ptosis are common initial complaints. Weakness in chewing is noticeable after prolonged effort. Speech may be affected secondary to weakness of the palate or tongue. Swallowing may result from weakness of the palate, tongue, or pharynx. In the majority of patients the weakness becomes generalized. The diagnosis is suspected after the appearance of the characteristic symptoms and signs. Edrophonium is an acetylcholinesterase inhibitor that allows ACh to interact repeatedly with the limited number of AChRs, producing improvement in the strength of myasthenic muscles. False-positive tests may occur in patients with other neurologic diseases. Electrodiagnostic testing may show evidence of reduction in the amplitude of the evoked muscle action potentials with repeated stimulation. Testing for the specific antibodies to AChR are diagnostic. In addition to anti-AChR antibodies, antibodies to MuSK have been found in some patients with clinical MG. Antibodies to voltage-gated calcium channels are found in patients with the Lambert-Eaton syndrome.

92. The answer is C.

(Chap. 47) Except for lumbar puncture, all of the options listed are indicated at this time. Thymic abnormalities are present in 75% of patients with myasthenia gravis. A CT or MRI of the mediastinum may show enlargement or neoplastic changes in the thymus and is recommended upon diagnosis. Hyperthyroidism occurs in 3–8% of patients with myasthenia gravis and may aggravate weakness. Testing for rheumatoid factor and antinuclear antibodies should also be obtained because of the association of myasthenia gravis to other autoimmune diseases. Due to side effects of immunosuppressive therapy, a thorough evaluation should be undertaken to rule out latent or chronic infections such as tuberculosis. Measurements of ventilatory function are valuable as a baseline because of the frequency and seriousness of respiratory impairment in myasthenic patients, and they can be used as an objective measure of response to therapy.

93. The answer is E.

(Chap. 48) All classes of lipid-lowering agents have been implicated in muscle toxicity including fibrates, HMG-CoA reductase inhibitors, niacin, and ezetimibe. Myalgia, malaise, and muscle tenderness are the most common manifestations, and muscle pain may be exacerbated by exercise. Proximal weakness may be found on examination. In severe cases, rhabdomyolysis and myoglobinuria may occur, though most cases are mild. Concomitant use of statins with fibrates and cyclosporine are more likely to cause adverse muscle reactions. Elevated serum CK is often identified, and muscle weakness is evidenced by myopathic EMG studies and myonecrosis on muscle biopsy. Severe myalgias, muscle weakness, and significant elevations in CK ($>3 \times$ upper limit of normal) and myoglobinuria are indications for stopping. After cessation, improvement generally occurs after several weeks.

94. The answer is E.

(Chap. 48) A number of endocrinologic conditions are associated with myopathy. Both hypo- and hyperthyroidism are associated with proximal muscle weakness. Hypothyroidism is frequently associated with an elevated CK, even with minimal clinical evidence of muscle disease. Thyrotoxic patients may have fasciculations in addition to proximal myopathy, but in contrast to hypothyroid patients, CK is not generally elevated. Hyperparathyroidism is associated with muscle weakness that is generally proximal. Muscle wasting and brisk reflexes are also generally present. Serum CK levels may be normal or slightly elevated. Serum calcium and phosphate levels show no correlation with clinical weakness. Hypoparathyroid patients also often have myopathy due to hypocalcemia. Patients with acromegaly usually have mild proximal weakness without atrophy. The duration of acromegaly, not the serum growth hormone levels, correlate with the degree of myopathy. Diabetes mellitus is a very rare cause of myopathy,

generally due to ischemic infarction of muscle and not a primary myopathy. Finally, vitamin D deficiency is associated with muscle weakness, as are glucocorticoid excess states, e.g., Cushing's disease.

95. The answer is D.

(Chap. 48) There are two recognized clinical forms of myotonic dystrophy, both of which are characterized by autosomal dominant inheritance. Myotonic dystrophy 1 (DM1) is the most common form and the most likely disorder in this patient. Characteristic clinical features of this disorder include a "hatchet-faced" appearance, due to wasting of the facial muscles, and weakness of the neck muscles. In contrast to the muscular dystrophies (Becker and Duchenne), distal limb muscle weakness is more common in DM1. Palatal, pharyngeal, and tongue involvement are also common and produce the dysarthric voice that is frequently heard. The failure of relaxation after a forced hand-grip is characteristic of myotonia. Percussion of the thenar eminence can also elicit myotonia. In most individuals, myotonia is present by age 5, but clinical symptoms of weakness that lead to diagnosis may not be present until adulthood. Cardiac conduction abnormalities and heart failure are also common in myotonic dystrophy. Diagnosis can often be made from clinical features alone in an individual with classic symptoms and a positive family history. An electromyogram would confirm myotonia. Genetic testing for DM1 would show a characteristic trinucleotide repeat on chromosome 19. Genetic anticipation occurs with an increasing number of repeats and worsening clinical disease over successive generations. Myotonic dystrophy 2 (DM2) causes primarily proximal muscle weakness and is also known by the name proximal myotonic myopathy (PROMM). Other features of the disease overlap with DM1. Acid maltase deficiency (glucosidase deficiency, or Pompe's disease) has three recognized forms, only one of which has onset in adulthood. In the adult-onset form, respiratory muscle weakness is prominent and often is the presenting symptom. As stated previously, Becker and Duchenne muscular dystrophies present with primarily proximal muscle weakness and are X-linked recessive disorders. Becker muscular dystrophy presents at a later age than Duchenne muscular dystrophy and has a more prolonged course. Otherwise, their features are similar. Nemaline myopathy is a heterogeneous disorder marked by the threadlike appearance of muscle fibers on biopsy. Nemaline myopathy usually presents in childhood and includes a striking facial appearance similar to that in myotonic dystrophy with a long, narrow face. This disease is inherited in an autosomal dominant fashion.

96. The answer is B.

(Chap. 49) When patients present with proximal muscle weakness and myositis, whether polymyositis, dermatomyositis, or inclusion body myositis, the diagnosis is

confirmed by analysis of serum muscle enzymes, EMG findings, and muscle biopsy. The most sensitive serum enzyme is creatinine kinase (CK), which can be elevated as much as 50-fold in active disease. CK levels usually parallel disease activity, but can be normal in some patients with inclusion body myositis or dermatomyositis. CK is always elevated in active polymyositis and thus is considered most sensitive. Other enzymes may be elevated as well including glutamic-oxaloacetic transaminase, glutamate pyruvate transaminase, lactate dehydrogenase, and aldolase.

97. The answer is B.

(Chap. 49) Various autoantibodies against nuclear antigens, e.g., ANAs, and cytoplasmic antigens are found in up to 20% of patients with inflammatory myopathies. The antibodies to cytoplasmic antigens are directed against ribonucleoproteins involved in protein synthesis (antisynthetases) or translational transport (anti-signal-recognition particles). The antibody directed against the histidyl-transfer RNA synthetase, called anti-Jo-1, accounts for 75% of all the antisynthetases and is clinically useful because up to 80% of patients with this autoantibody will have interstitial lung disease. Patients with anti-Jo-1 may also have Raynaud's phenomenon, nonerosive arthritis, and the MHC molecules DR3 and DRw52. Interstitial lung disease associated with anti-Jo-1 is often rapidly progressive and fatal, even if treated aggressively with cyclophosphamide or other immunosuppressants.

98. The answer is A.

(Chap. 49) Dermatomyositis is associated with malignancy in up to 15% of cases, thus age-appropriate cancer screening is indicated when this diagnosis is made. Exhaustive cancer searches are not recommended, however. Dermatomyositis may be associated occasionally with scleroderma and mixed connective tissue disease, but less frequently with systemic lupus erythematosus, rheumatoid arthritis, or Sjögren's syndrome, which are more closely associated with polymyositis or inclusion body myositis (IBM). Viruses may be associated with IBM and polymyositis, but are not proven to be associated with dermatomyositis. Parasites and bacteria such as cestodes and nematodes are associated with polymyositis, but not other forms of inflammatory myopathy. Finally, thyroid-stimulating immunoglobulins are not known to be associated with dermatomyositis.

99. The answer is A.

(Chap. 49) A common mistake in the management of patients with inflammatory myopathy is to "chase the CK" instead of adjusting therapy based on the clinical response. The goal of therapy is to improve strength. If that goal is being achieved, no augmentation of therapy is necessary. In this case, the plan to switch to long-term maintenance with steroid-sparing immunosuppressants should still be

pursued. There have been no controlled studies comparing mycophenolate to methotrexate for long-term use in polymyositis, and in the absence of an adverse reaction to mycophenolate, therapy should not be changed. Despite an elevated CK, patients with polymyositis who are responding to therapy do not need a repeat muscle biopsy.

100. The answer is B.

(Chap. 50) The FLAIR MRI shows increased signal bilaterally in the occipital lobes predominantly involving the white matter. This pattern is typical of a hyperperfusion state, in this case secondary to calcineurin-inhibitor toxicity. This clinical-radiographic abnormality was previously described as reversible posterior leukoencephalopathy. However, this characterization is no longer utilized because the syndrome may not be reversible, the territory may not be confined posteriorly, and gray matter may be involved. Hyperperfusion syndrome may be due to a hydrostatic elevation in cerebral capillary pressure or disorders with endothelial dysfunction and capillary leakage. Hydrostatic causes include hypertensive encephalopathy, post-carotid endarterectomy syndrome, (pre)eclampsia, and high-altitude cerebral edema. Endothelial dysfunction causes include calcineurin-inhibitor toxicity, other chemotherapeutic agent toxicity, HELLP syndrome, TTP, SLE, or granulomatosis with polyangiitis (Wegener's). The diagnosis is clinical and radiographic. CSF findings are nonspecific. In the case of cyclosporine the syndrome may occur with therapeutic serum levels and sporadically after years of treatment. Acoustic neuroma would present on MRI with a discrete mass. The radiologic and clinical appearance is not consistent with pituitary apoplexy. Post-liver transplant patients are at risk of acute (streptococcal) and chronic (tuberculous) meningitis, but the clinical and radiologic findings in this case would not be typical.

101. The answer is B.

(Chap. 50) Acute neurologic events, such as encephalopathy or stroke (often related to intraoperative hypotension or embolism), are common after open heart surgery or coronary artery bypass graft (CABG). Additionally, a chronic syndrome of cognitive impairment is now increasingly recognized after surgery. Small or microemboli during surgery are thought to be the etiology of a hyper- or hypoactive confusional state in the postoperative period. A smaller burden of microemboli may be responsible for the more subtle post-cardiac surgery syndrome characterized by confusion and depressive symptoms as described in this case. Cardiac surgery may also unmask the early manifestations of vascular dementia or Alzheimer's disease. Off-pump CABG patients have a shorter length of hospital stay and fewer perioperative complications. Recent studies do not confirm the hypothesis that off-pump surgery results in less cognitive impairment than on-pump surgery. Ongoing studies are testing the efficacy of microfilters to capture emboli and

reduce CNS complications. Given the temporal relation to CABG surgery and the lack of temporal variability, multiple sclerosis is not likely in this case. Similarly, in the absence of meningitis or encephalitis signs or symptoms, streptococcal or West Nile virus disease is unlikely. vCJD is associated with ingestion of prion-contaminated product. It is characterized by rapidly developing delirium and dementia, often associated with myoclonic jerks.

102. The answer is E.

(Chap. 50) The peroneal nerve winds around the head of the fibula below the lateral aspect of the knee. This superficial location makes it vulnerable to trauma. Poorly applied leg braces, fibular fracture, tight-fitting stockings, or casts may cause peroneal nerve injury and neuropathy. Patients may present with footdrop (dorsiflexion defect) with weakness of foot eversion. Intact foot inversion at the ankle distinguishes peroneal nerve injury from L5 radiculopathy, which involves the muscles innervated by the tibial nerve. Sensory loss due to peroneal nerve injury involves the lateral aspect of the leg below the knee and the dorsum of the foot. Cauda equina syndrome is caused by compression of the spinal nerve roots of the lumbar plexus usually due to tumor, trauma, or spinal stenosis. It typically presents with weakness of the muscles innervated by the involved nerves, incontinence, and decreased anal sphincter tone. Cauda equina syndrome is usually a medical emergency to avoid permanent functional loss. The femoral nerve branches into anterior and posterior portions in the leg. Injury of the anterior branches may lead to sensory findings in the thigh and muscular findings in the sartorius and the quadriceps. L4 radiculopathy causes symptoms in the anterior thigh and knee extensors (including the patellar reflex).

103. The answer is A.

(Chap. 52) Chronic fatigue syndrome (CFS) is a disorder characterized by persistent and unexplained fatigue resulting in severe impairment in daily functioning. Besides intense fatigue, most patients with CFS report concomitant symptoms such as pain, cognitive dysfunction, and unrefreshing sleep. Additional symptoms can include headache, sore throat, tender lymph nodes, muscle aches, joint aches, feverishness, difficulty sleeping, psychiatric problems, allergies, and abdominal cramps. Criteria for the diagnosis of CFS have been developed by the U.S. Centers for Disease Control and Prevention (see Table 52-1). CFS is seen worldwide, with adult prevalence rates varying between 0.2% and 0.4%. In the United States, the prevalence is higher in women, members of minority groups (African and Native Americans), and individuals with lower levels of education and occupational status. Approximately 75% of all CFS patients are women. The mean age of onset is between 29 and 35 years. It is probable that many patients go undiagnosed and/or do not seek help.

104. The answer is B.

(Chap. 52) Cognitive behavioral therapy (CBT) and graded exercise therapy (GET) have been found to be the only beneficial interventions in chronic fatigue syndrome (CFS). CBT is a psychotherapeutic approach directed at changing condition-related cognitions and behaviors. CBT for CFS aims at changing a patient's perpetuating factors by exploiting various techniques and components. The intervention, which typically consists of 12–14 sessions spread over 6 months, helps CFS patients gain control over their symptoms. GET is based on the model of deconditioning and exercise intolerance and usually involves a home exercise program that continues for 3–5 months. Walking or cycling is systematically increased, with set target heart rates. Evidence that deconditioning is the basis for symptoms in CFS is lacking, however. The primary component of CBT and GET that results in a reduction in fatigue is a change in the patient's perception of fatigue and focus on symptoms. CBT is generally the more complex treatment, which might explain why CBT studies tend to yield better improvement rates than GET trials. Not all patients benefit from CBT or GET. Predictors of poor outcome are somatic comorbidity, current disability claims, and severe pain. CBT offered in an early stage of the illness reduces the burden of CFS for the patient as well as society in terms of decreased medical and disability-related costs. Full recovery from untreated CFS is rare: the median annual recovery rate is 5% (range 0–31%) and the improvement rate 39% (range 8–63%). Major depressive disorder, bipolar disorder, eating disorder, and schizophrenia are exclusion criteria for the diagnosis of chronic fatigue syndrome.

105. The answer is D.

(Chap. 54) This patient is experiencing her first episode of a panic attack and does not meet the criteria for panic disorder. In this situation, no specific treatment is required. The patient should be reassured in a manner that is empathetic and supportive that she does not have any evidence of a serious medical disorder. Panic attacks are common, with about 1–3% of the population experiencing at least one panic attack. Panic attacks begin abruptly, most commonly without an immediate precipitating cause, and peak in severity over 10 minutes. The symptoms usually subside spontaneously over the course of an hour. Diagnostic criteria for a panic attack include a minimum of four of the following criteria: palpitations or racing heart, sweating, trembling, shortness of breath, feeling of choking, chest pain, nausea or GI distress, dizziness, derealization, fear of losing control, fear of dying, paresthesias, or chills/hot flushes. If a patient subsequently develops panic disorder, a variety of treatment options can be pursued. Panic disorder is marked by at least 1 month of recurrent panic attacks associated with excessive worry about or change in behavior as a result of the attacks. The goals of therapy for panic attacks are to decrease the frequency of attacks and

the severity of symptoms during the attack. Antidepressant medications are the cornerstone of therapy with selective serotonin reuptake inhibitors being the most frequently used class of medication. The dose of medication for panic disorder is typically lower than the antidepressant dose. For fluoxetine, this would be 5–10 mg daily. As these medications take 2–6 weeks to become effective, they are often combined with benzodiazepines to be used on an as-needed basis for immediate relief of attacks. Alprazolam and clonazepam are common agents used for panic disorder, although alprazolam may have more associated dependence with the need for escalating doses of medications. In combination with pharmacologic therapy, psychotherapy and education are also useful for the treatment of panic disorder. The therapy often includes breathing techniques, cognitive behavioral therapy, and even homework assignments.

106. The answer is A.

(Chap. 54) There are increasing numbers of antidepressant medications available in a variety of classes. Selective serotonin reuptake inhibitors (SSRIs) are the most commonly used antidepressant drugs. This class of medications includes fluoxetine, sertraline, paroxetine, fluvoxamine, citalopram, and escitalopram. These medications are taken once daily and have side effects including sexual dysfunction, headache, and insomnia. Tricyclic antidepressants were commonly used in past decades for the treatment of depression. However, overdoses can be lethal, and anticholinergic side effects including dry mouth, constipation, and urinary retention can limit the dose. Medications in the tricyclic class of antidepressants include amitriptyline, nortriptyline, imipramine, desipramine, doxepin, and clomipramine. Mixed norepinephrine/serotonin reuptake inhibitors and receptor blockers are a newer class of medications. These medications are increasing in use as they are quite effective and do not have the same frequency of sexual dysfunction. Medications in this class include venlafaxine, desvenlafaxine, duloxetine, and mirtazapine. Monoamine oxidase inhibitors were once a common antidepressant class of medication, but these medications are now only rarely used. There is a wide range of drug and food interactions that can lead to hypertensive crises. Examples of medication in this class include phenelzine, tranylcypromine, and isocarboxazid. A final class of antidepressants is called simply mixed action drugs and includes trazodone, bupropion, and nefazodone.

107. The answer is E.

(Chap. 54) Post-traumatic stress disorder (PTSD) was only added as a discrete disorder in 1980. The diagnostic criteria for PTSD are long and require that an individual experiences an event where there was an actual or perceived threat of death or serious injury and that the individual's reaction included intense fear or help-

lessness. Following the event, the individual continues to re-experience the event and avoids stimuli associated with the trauma. In association with this, there is also often a generalized withdrawal and decrease in responsiveness. At the same time, the patient exhibits an increase in arousal that is often exhibited by insomnia, irritability, hypervigilance, and difficulty concentrating. Treatment of PTSD is almost always multifactorial, including both pharmacotherapy and psychotherapy. It is not uncommon for an individual with PTSD to develop a dependence on drugs or alcohol as an attempt to control the symptoms, and any substance abuse issues need to be treated simultaneously as well. This patient's treatment would include avoidance of alcohol and intensive substance abuse treatment as needed. Treatment with antidepressant medications can decrease anxiety and avoidance behaviors. Trazodone is often given at night for its sedating properties. Psychotherapeutic strategies include cognitive behavioral therapy to overcome avoidance behaviors as well.

108. The answer is B.

(Chap. 54) Fifteen percent of the population will experience at least one episode of major depression over the course of a lifetime, and most episodes of major depression are treated by primary care practitioners. Treatment can be any of a number of medications across a variety of classes. Despite the popularity of newer antidepressants, there is no evidence that these medications are more efficacious than older drugs like tricyclic antidepressants. Indeed, 60–70% of patients will respond to any drug chosen if given in a sufficient dose for 6–8 weeks. Once a patient has been on treatment for about 2 months, the response should be evaluated, and if there has been an insufficient response, a dosage increase should be considered. In this patient, a dosage increase yielded control of depressive symptoms at 4 months. Once control of symptoms has been achieved, the drug should be continued for an additional 6–9 months to prevent relapse. If a patient experiences any additional episodes of major depression, he or she will likely require indefinite maintenance treatment.

109. The answer is A.

(Chap. 56) Alcohol is primarily absorbed through the proximal small intestine, but small to moderate amounts can also be absorbed in the mouth, esophagus, stomach, and large intestines. Several factors can increase the rate of absorption. One factor that increases absorption is rapid gastric emptying, which can be induced by concurrent consumption of carbonated beverages. Another factor that increases absorption from the gut to the blood is the ingestion of alcohol in the absence of other calorie sources such as proteins, fat, or carbohydrates. A final factor that can increase absorption is to drink alcohol that is diluted to a modest concentration (~20% or less). At high

alcohol concentrations absorption is decreased, although high blood levels may be achieved because the amount of alcohol ingested is high.

110. The answer is C.

(Chap. 56) Alcohol has effects on many neurotransmitters in the brain. The predominant effect of alcohol lies in its ability to cause the release of γ -aminobutyric acid (GABA) and acts primarily at the GABA_A receptors. GABA is the primary inhibitory neurotransmitter in the brain and is associated with the sedative effects of alcohol. Many other drugs affect the GABA system including benzodiazepines, nonbenzodiazepine sleep aids such as zolpidem, anticonvulsants, and muscle relaxants. The euphoric effects of alcohol consumption are related to increases in dopamine, which is common to all pleasurable activities. The effects on dopamine are thought to be important in alcohol craving and relapse. In addition, alcohol alters opioid receptors and can lead to a release of beta endorphins during acute ingestion. In addition to these effects, alcohol also inhibits postsynaptic *N*-methyl-D-aspartate excitatory glutamate receptors. Glutamate is the primary excitatory neurotransmitter of the brain, and its inhibition further contributes to the sedative effects of alcohol. Additional important effects on neurotransmitters include increased serotonin activity and decreased nicotinic acetylcholine receptors.

111. The answer is D.

(Chap. 56) The acute effects of any drug depend on many factors including amount consumed and absorbed, presence of other drugs, and past experience with the drug. In an individual who is naïve to alcohol, drug levels as low as 0.02 g/dL can lead to a decrease in inhibitions and a slight feeling of intoxication. In the United States, “legal” intoxication occurs at a blood alcohol level of 0.08 g/dL in most states. At this level, decreases in cognitive and motor abilities are seen. Once an alcohol level of 0.20 g/dL is achieved, an individual is obviously impaired with slurred speech, poor judgment, and impaired coordination. Light coma and depression of respiratory rate, blood pressure, and pulse occur at levels of around 0.30 g/dL, and death is likely to occur at levels of 0.40 g/dL. However, in individuals who drink heavily, tolerance begins to develop to alcohol. After a period of 1–2 weeks of daily alcohol consumption, liver metabolism of alcohol increases by as much as 30%, but disappears quite quickly with abstinence. Cellular or pharmacodynamic tolerance also occurs and refers to the neurochemical changes that allow an individual to maintain more normal physiologic functioning despite the presence of alcohol.

112. The answer is C.

(Chap. 56) Alcohol abuse is defined as repetitive problems in any one of four life areas that can be attributed to alcohol. The four life areas that can be affected by alco-

hol include social, interpersonal, legal, or occupational. In addition, an individual who repetitively engages in hazardous behaviors while under the influence of alcohol would be considered to suffer from alcohol abuse. However, this is to be differentiated from alcohol dependence. Alcohol dependence is defined in the DSM-IV as repeated alcohol-related difficulties in three of seven life areas and includes the development of tolerance and dependence. If tolerance or dependence is present, this predicts a more severe clinical course, and the presence of alcohol dependence decreases overall life span by about a decade. Only about 50% of individuals with alcohol abuse will continue to experience similar alcohol-related problems 3–5 years later, and only 10% will go on to develop alcohol dependence. The lifetime risk of alcohol dependence in most Western countries is about 10–15% in men and 5–8% in women. However, there may be higher rates in Ireland, France, and Scandinavian countries. In addition, native cultures appear to be especially susceptible to problems with alcohol dependence. This has been seen in Native Americans, Maoris, and the aboriginal tribes of Australia.

About 60% of the risk for alcohol use disorders is attributed to genetic influences. Children of alcoholics do have a higher risk of alcohol abuse and dependence; however, this risk is about 4 times higher, not 10. Identical twins also exhibit a higher risk of concurrent alcohol abuse and dependence when compared to fraternal twins. The genetic factors that appear to be most strongly linked to alcohol use disorders include genes that are linked to impulsivity, schizophrenia, and bipolar disorder. In addition, genes that affect alcohol metabolism or sensitivity to alcohol also contribute to the genetics of alcoholism. A mutation in aldehyde dehydrogenase that is more common in individuals of Asian descent results in intense flushing when alcohol is consumed and confers a decreased risk of alcohol dependence. Conversely, genetic variants that lead to a low sensitivity to alcohol increase the risk of subsequent alcohol abuse and dependence, as higher and higher doses of alcohol are required to achieve the same effects.

113. The answer is B.

(Chap. 56) Individuals with alcohol dependence are susceptible to alcohol withdrawal when alcohol intake is stopped abruptly. The individual in this case scenario is likely alcohol dependent given his large amount of alcohol intake on a daily basis. Symptoms of alcohol withdrawal can range from mild tremulousness to hallucinations, seizures, or the development of delirium tremens. Other clinical features of alcohol withdrawal include anxiety, insomnia, and autonomic nervous system overactivity manifested as tachycardia, tachypnea, elevated blood pressure, and fever. This patient exhibits symptoms of the more severe delirium tremens, with mental confusion, agitation, and fluctuating levels of consciousness. While minor symptoms of alcohol withdrawal may begin

as soon as 5–10 hours after cessation of alcohol intake, the symptoms do not peak for 48–72 hours, putting this patient in the appropriate time frame for alcohol withdrawal.

The best approach for the alcohol-dependent patient who abruptly stops all alcohol intake is to take a prophylactic approach and screen early for symptoms of alcohol withdrawal. Tools such as the Revised Clinical Institute for Withdrawal Assessment for Alcohol (CIWA-Ar) may help clinicians and nurses screen for the early development of symptoms and allow intervention before symptoms escalate. In this setting, most experts recommend the use of oral long-acting benzodiazepines such as chlordiazepoxide or diazepam beginning on the first day. However, in this case, the patient received no such treatment and is now experiencing severe alcohol withdrawal and delirium tremens. Intravenous medications that have a rapid onset of action and can be titrated for more aggressive symptom management are often employed in this setting. Thus, the use of IV lorazepam or diazepam is preferred in this patient. Following an initial bolus, repeated doses can be used in short intervals until the patient is calm but arousable. In some instances a continuous infusion may be required, although bolus dosing is preferred. In the most severe cases, propofol or barbiturates may be required, although the patient would most likely need to be intubated for airway protection with use of these medications.

The other options listed are not appropriate for initial management of this patient. Intravenous fluids and thiamine had been administered since hospital admission. Administration of glucose-containing fluids without thiamine in the alcohol-dependent patient can precipitate Wernicke's encephalopathy, which would present with ophthalmoparesis, ataxia, and encephalopathy. Given the patient's fever, an infectious etiology can be considered, and it would be appropriate to perform blood cultures in this patient. However, given the clear symptoms of alcohol withdrawal and lack of necrotizing pancreatitis on CT abdomen, empiric treatment with antibiotics is not required. Likewise, without focal neurologic findings, a head CT would be a low-yield diagnostic procedure that would be difficult to perform in the patient's current agitated condition and would only delay appropriate therapy. Finally, restraints are best avoided if the patient's safety can be ensured through the appropriate use of benzodiazepines, as restraints are only likely to make the patient's agitation worse and may lead to iatrogenic harm. Haloperidol may have some sedative effect on the patient, but could lead to torsades de pointe arrhythmia as this patient is at risk for electrolyte deficiencies from his alcoholism and pancreatitis.

114. The answer is D.

(Chap. 56) In individuals recovering from alcoholism several medications may have a modest benefit in increasing abstinence rates. The two medications with the best

risk-benefit ratio are acamprosate and naltrexone. Acamprosate inhibits NMDA receptors, decreasing symptoms of prolonged alcohol withdrawal. Naltrexone is an opioid antagonist that can be administered orally or as a monthly injection. It is thought to act by decreasing activity in the dopamine-rich ventral tegmental area of the brainstem and subsequently decreasing the pleasurable feelings associated with alcohol consumption. There is some research to suggest that the use of these medications in combination may be more effective than either one alone. Disulfiram is an aldehyde dehydrogenase inhibitor that has been used for many years in the treatment of alcoholism. However, it is no longer a commonly used drug due to its many side effects and risks associated with treatment. The primary mechanism by which it acts is to create negative effects of vomiting and autonomic nervous system hyperactivity when alcohol is consumed concurrently with use of the medication. As it inhibits an enzyme that is part of the normal metabolism of alcohol, it allows the buildup of acetaldehyde, which creates these symptoms. Because of the autonomic side effects, it is contraindicated in individuals with hypertension, a history of stroke, heart disease, or diabetes mellitus.

115. The answer is E.

(Chap. 57) Prescription drug abuse has increased dramatically among all age groups and is strikingly common in teenagers. Since 2007, prescription opiates have passed marijuana as the most common illicit drugs that adolescents initially abuse. This has occurred at the same time as rates of prescription narcotic abuse have increased across all age groups. The annual prevalence of heroin abuse is approximately 0.14% of the population. In contrast, this prevalence is only one-third the rate of prescription opiate abuse. Among prescription narcotics, oxycodone is the single most commonly abused drug. Other common prescription narcotics that are abused include morphine and hydrocodone. Among health care professionals, meperidine and fentanyl are more frequently abused.

116. The answer is B.

(Chap. 57) Tolerance and withdrawal begin within 6–8 weeks of chronic daily opioid use. Tolerance develops not because of increased metabolism, but through a change in the pharmacodynamics of the drugs, requiring increasing doses to achieve the euphoric effects and prevent withdrawal. With the abrupt cessation of narcotics, acute withdrawal symptoms begin within 8–10 hours after the last dose. While the symptoms of narcotic withdrawal are noxious, they are not life threatening, as is the case with benzodiazepine or barbiturate withdrawal. The primary symptoms of opiate withdrawal are related to overactivity of the autonomic nervous system. This manifests as increased lacrimation, rhinorrhea, and sweating. In addition, patients frequently will have diffuse piloerection (chill bumps), giving rise to the term “cold turkey.” As

withdrawal symptoms progress, patients appear restless with myalgias, nausea, vomiting, and diarrhea. Hypertension, hyperthermia, and tachypnea can occur as well. Hypotension is not a symptom of opioid withdrawal. A patient with known infection and new-onset hypotension should be evaluated for systemic infection, not withdrawal.

117. The answer is E.

(Chap. 57) The patient is presenting with an acute overdose of an unknown quantity of extended-release opioid medications taken with alcohol. When evaluating and treating a patient with an intention overdose, the first priority is to stabilize the patient's condition. The patient was appropriately given the opiate antagonist naloxone by emergency responders as the patient was near-apneic. In addition, the patient was also appropriately intubated and stabilized for transport to the hospital. In the emergency room, however, the patient remained hypotensive and unresponsive. At this point, the next step in stabilizing the patient is to support the blood pressure with bolus fluid resuscitation, and if the patient fails to respond, IV vasopressors would be required. Given his ongoing unresponsive state and the expected long duration of effect with a sustained-release preparation, it is appropriate to initiate a continuous infusion of naloxone. After a bolus dose, the expected onset of action is 1–2 minutes, but the duration of effect is only a few hours. Some care must be taken when giving naloxone as one only wants to reverse the respiratory and cardiovascular depression associated with the overdose. Particularly in chronic drug abusers, high doses of naloxone can precipitate the distressing symptoms of narcotic withdrawal. When long-acting preparations of opioids are taken, activated charcoal and gastric lavage are appropriate considerations to decrease the absorption of any undigested pills. While the patient is being stabilized from a cardiovascular and respiratory standpoint, it is important that the clinician consider if any other concurrent ingestion may have occurred that would affect the patient's outcome. As the patient is unable to provide any history and the overdose was not witnessed, one must not focus solely on the opioids. The appropriate approach is to perform a comprehensive toxicology evaluation that should include a urine drug screen, blood alcohol level, and acetaminophen levels, at a minimum. One could also consider sending for levels of aspirin or tricyclic antidepressants.

118. The answer is C.

(Chap. 58) Marijuana is the most commonly used illegal drug in the United States with over 6% of all in-

dividuals reporting current usage in 2009 (<http://www.whitehousedrugpolicy.gov/publications/pdf/nsduh.pdf>, accessed July 25, 2011). In part, the prevalence of marijuana use is related to the widespread belief that marijuana is thought to have few negative health effects. Acutely, marijuana causes a sense of relaxation and mild euphoria, not unlike that of alcohol intoxication. In addition, impaired judgment, cognition, and psychomotor performance are seen. Occasionally, acute intoxication can lead to negative emotional responses as well. A consensus about the chronic effects of marijuana usage is not clearly defined. Traditionally, marijuana usage has been linked with an "amotivational syndrome." While it is true that chronic users of marijuana may lose interest in day-to-day activities and spend more time using the drug, this is certainly not specific for marijuana uses, and a specific "amotivational syndrome" is not defined with chronic marijuana use. Other symptoms that have been attributed to chronic marijuana use that lack good evidence for causation include depression and maturational dysfunction. In individuals with a history of schizophrenia, however, chronic marijuana use has been associated with an increased risk of psychotic symptoms.

The physical effects of chronic marijuana use are also not clearly known. Acutely, marijuana causes increased heart rate, but tolerance for this effect occurs rapidly. Acute ingestion can also precipitate angina. The chronic effects on lung function are not known, as tobacco products are frequent confounders. Acute decreases in vital capacity and diffusion capacity are seen, but whether this translates into an increased risk of emphysema has not yet been determined. Most studies have not found an association with emphysema. A variety of other adverse physical effects have been described but not confirmed in a systematic fashion. This includes reports of low testosterone levels, decreased sperm count, impaired fetal growth, and chromosomal abnormalities.

Contrary to popular belief, chronic use of marijuana is associated with the development of tolerance as well as a withdrawal syndrome. Signs of physical tolerance include tolerance to the development of tachycardia and conjunctival injection. The psychological tolerance that develops is more prominent and predictable. This occurs rapidly, with individuals often seeking more potent compounds or smoking the drug more frequently. With cessation of marijuana use, a withdrawal syndrome can be demonstrated with irritability, anorexia, and sleep disturbance.

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